

Faculty of Computer Science and Engineering
Ho Chi Minh City University of Technology

Chapter 3

Data Regression

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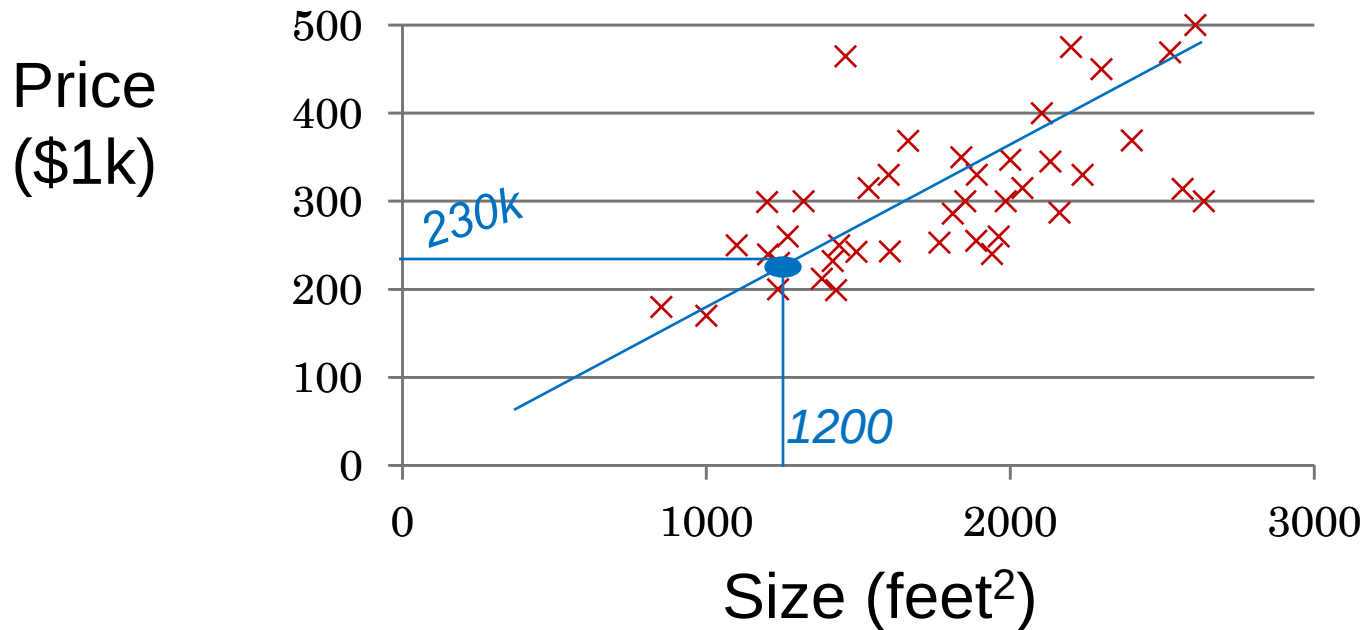
CONTENT

1. Introduction
2. Linear regression
3. Non-Linear regression
4. Applications
5. Problems with regression
6. Summary

REFERENCES

- [1a] Jiawei Han, Micheline Kamber, and Jian Pei, “Data Mining: Concepts and Techniques”, 3rd Edition, Morgan Kaufmann Publishers, 2012.
- [1b] Trần Minh Quang, "Khai Phá Dữ Liệu và Kỹ Thuật Phân Lớp", NXB Đại Học Quốc Gia TP. HCM, 2020.
- [2] David Hand, Heikki Mannila, Padhraic Smyth, “Principles of Data Mining”, MIT Press, 2001.
- [3] David L. Olson, Dursun Delen, “Advanced Data Mining Techniques”, Springer-Verlag, 2008.
- [4] Graham J. Williams, Simeon J. Simoff, “Data Mining: Theory, Methodology, Techniques, and Applications”, Springer-Verlag, 2006.
- [5] ZhaoHui Tang, Jamie MacLennan, “Data Mining with SQL Server 2005”, Wiley Publishing, 2005.
- [6] Oracle, “Data Mining Concepts”, B28129-01, 2008.
- [7] Oracle, “Data Mining Application Developer’s Guide”, B28131-01, 2008.
- [8] Ian H.Witten, Eibe Frank, “Data mining : practical machine learning tools and techniques”, 2nd Edition, Elsevier Inc, 2005.
- [9] Florent Messegli, Pascal Poncelet & Maguelonne Teisseire, “Successes and new directions in data mining”, IGI Global, 2008.

1. INTRODUCTION



+ Can we model the house price distribution based on their sizes ?

+ Can we predict a house price based on its size?

1. INTRODUCTION

Microsoft Excel - stb.csv

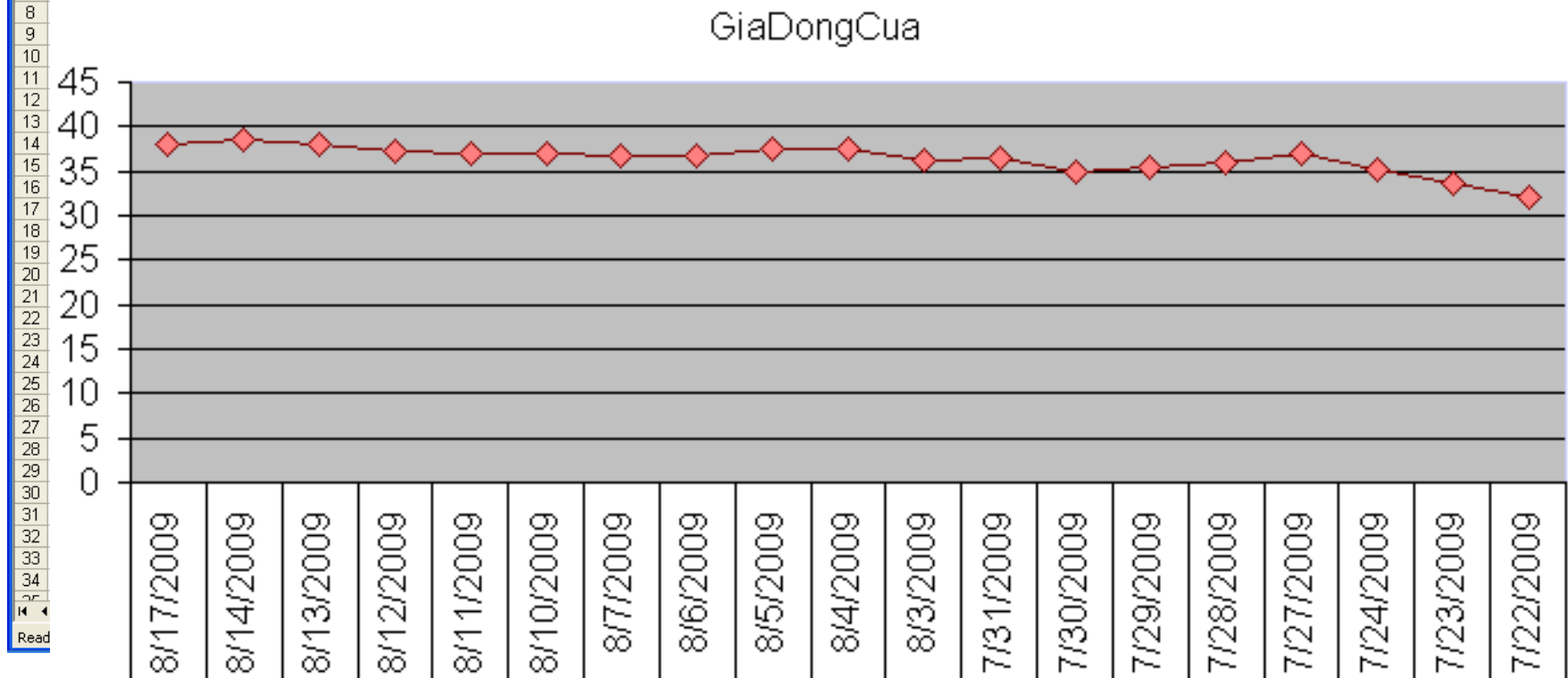
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A1 MaCK

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1	MaCK	Ngay	GiaMoCua	GiaCaoNhat	GiaThapNhat	GiaDongCua	KhoiLuongGD	GiaTran	GiaSan	GiaThamChieu	TangGiam	%	GDThoaThua
2	STB	8/17/2009	38.5	38.8	38.1	38.1	5986700	40.4	36.6	38.5	-0.4	-1.04	24343
3	STB	8/14/2009	38	38.7	38	38.5	6886430	39.9	36.1	38	0.5	1.32	340000
4	STB	8/13/2009	38	38.5	37.6	38	8716920	39	35.4	37.2	0.8	2.15	188000
5	STB	8/12/2009	37.3	37.4	37	37.2	5361890	38.7	35.1	36.9	0.3	0.81	200000
6	STB	8/11/2009	37.1	37.3	36.9	36.9	3675610	38.9	35.3	37.1	-0.2	-0.54	0
7	STR	8/10/2009	37.2	37.6	36.8	37.1	6140320	38.5	34.9	36.7	0.4	1.09	0

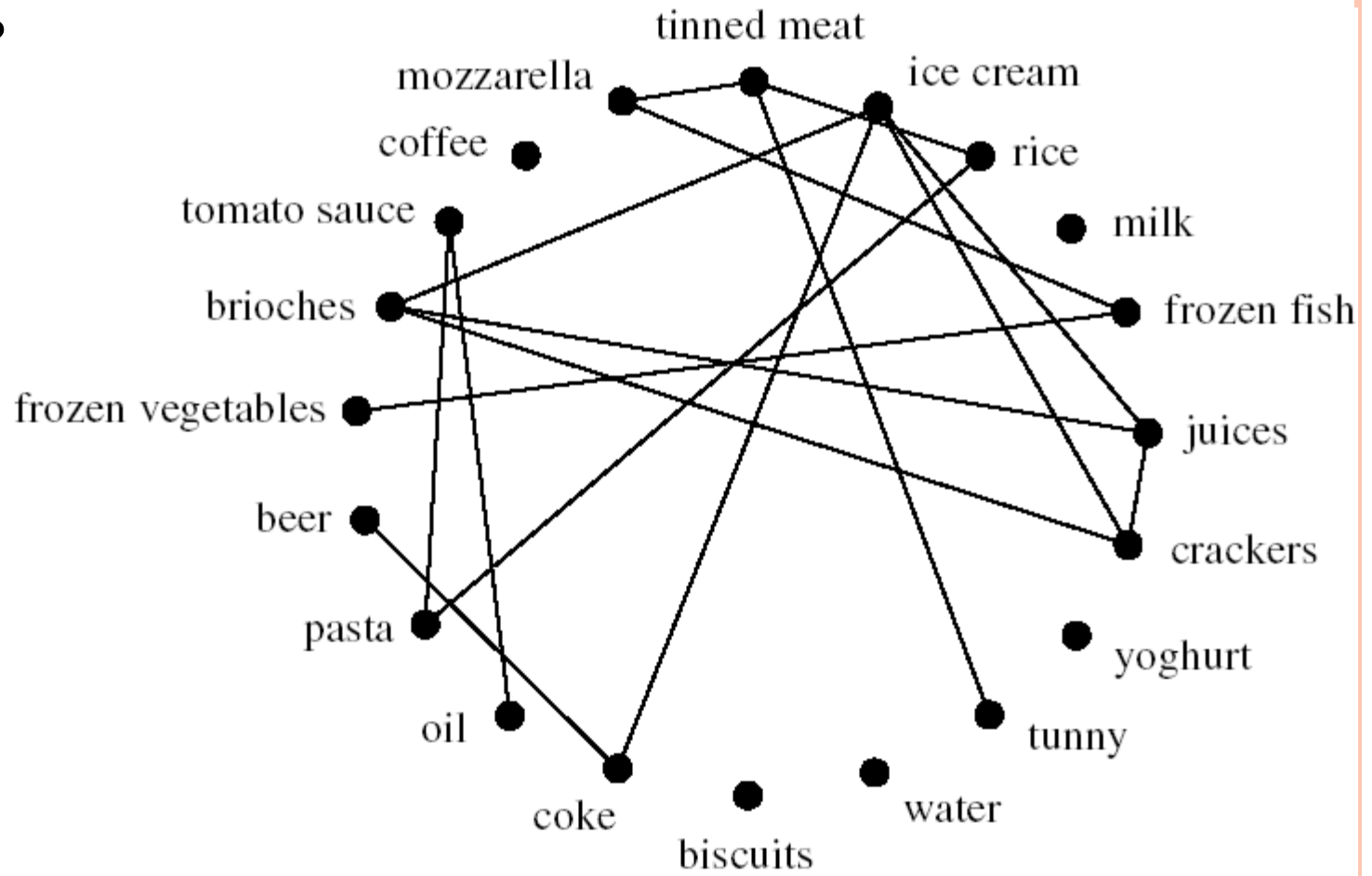
Can we predict a stock price using regression model?



1. INTRODUCTION

The market basket analysis problem

→ Can we find out association rules between products in transactions?



1. INTRODUCTION

- Analyzing the factors that impact on the quality of e-banking services (based on surveys from users)
 - Easy to use (+0.209)
 - Fast response (+0.261)
 - The ability to link with other billing services (+0.199)
 - Feelings of individuality (+0.15)
 - Privacy and security issues (-0.25)
 - ...

1. INTRODUCTION

○ Regression

- J. Han et al (2001, 2006): Regression is a statistic mechanism that allows predicting real/numeric and continuous values
- Wiki (2009): Regression analysis is a statistic mechanism that allows estimating the correlation between independent variables
- R. D. Snee (1977): Regression is a statistic mechanism in data analytics and building models from experiments, it allows prediction, control, and learning the rules to which data is generated.
- **Regression: Numeric data prediction (real-valued output)**
- **Classification: “prediction” for discrete values**

1. INTRODUCTION

- Regression model: Describe the relationship between a set of predictors/independent variables and one or some responses/dependent variables
- Regression equation

$$Y = f(X, \theta)$$

X: a set of predictors/independent variables; describes the changes of responses/dependent variables Y

Y: responses/dependent variables; Describes the interesting facts/events

θ : Regression coefficients; Describes the relative effects of X on Y

1. INTRODUCTION

○ Categories:

- Linear v.s nonlinear
 - ✓ Linear in parameters: Linear association between parameters that affect Y
 - ✓ Nonlinear in parameters: Non-linear association between parameters that affect Y
- Single variable v.s multiple variables
 - ✓ Single: $X = (X_1)$ v.s. Multiple: $X = (X_1, X_2, \dots, X_k)$
- Parametric v.s nonparametric and semiparametric
- Symmetric v.s asymmetric
 - ✓ Symmetric: descriptive regression models (e.g., log-linear models)
 - ✓ Asymmetric: predictive regression models (e.g., generalized linear models)

1. INTRODUCTION

- Parametric, nonparametric, and semiparametric
 - Parametric: regression models with finite parameters
 - Nonparametric: regression models with infinite parameters
 - Semiparametric: regression models with **finite interesting** parameters

Regression model□	Description□
Parametric□	$Y = \theta_0 + \theta_1 * X$
Nonparametric□	$Y = \theta_0 + f(X)$
Semiparametric□	$Y = \theta_0 + \theta_1 * X_1 + f(X_2)$

2. LINEAR REGRESSION

- Single variable (Univariate)
- Multiple variables (Multivariate)

2.1. UNIVARIATE LINEAR REGRESSION

○ Notations

- ✓ N : size of training examples
- ✓ x : input variable/feature
- ✓ y : output/target variable
- ✓ $(x^{(i)}, y^{(i)})$: i^{th} learning sample
- ✓ $(x^{(1)}, y^{(1)}) = (2100, 450)$

Size feet ² (x)	Price (\$1k) (y)
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2100	450
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1416	232
------	-----

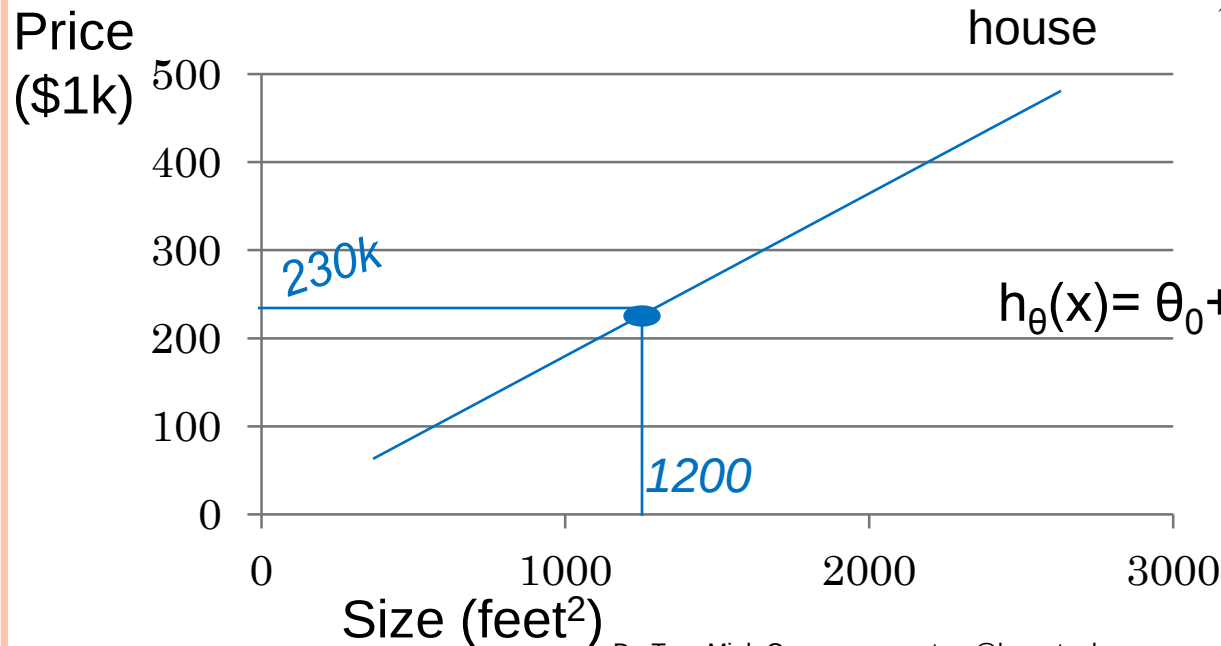
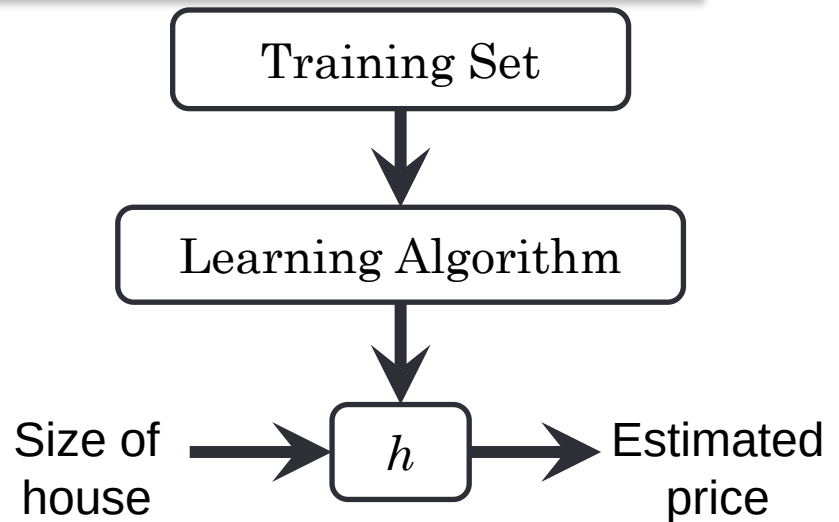
1534	315
------	-----

852	178
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...	...
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2.1. UNIVARIATE LINEAR REGRESSION

- Hypothesis (h)
- $y = h(x)$; h is a mapping from x to y
- How to model h ?



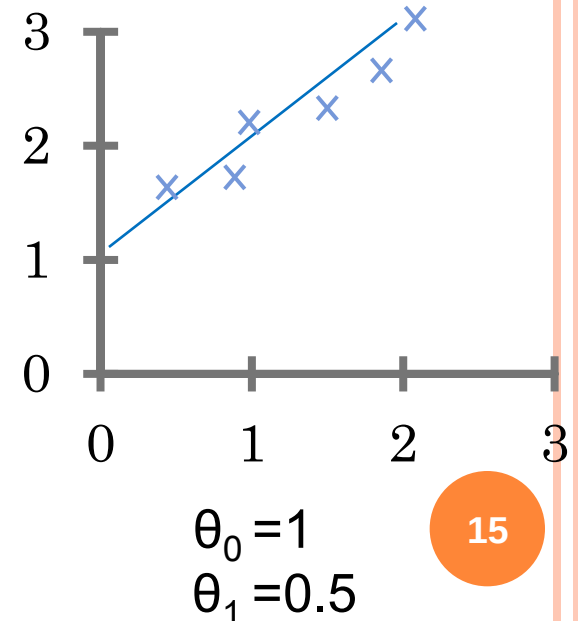
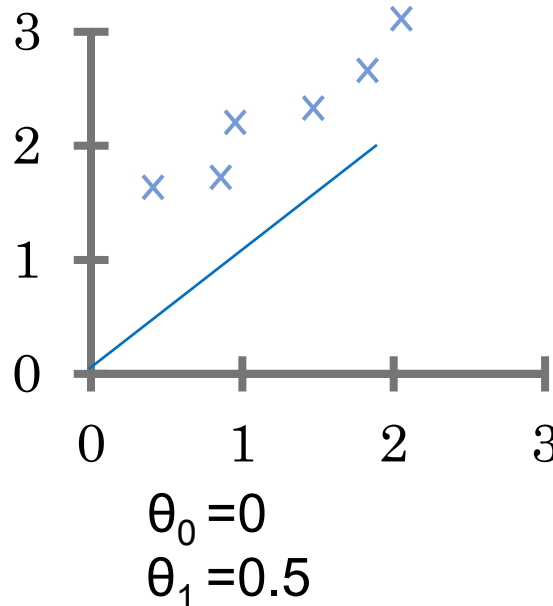
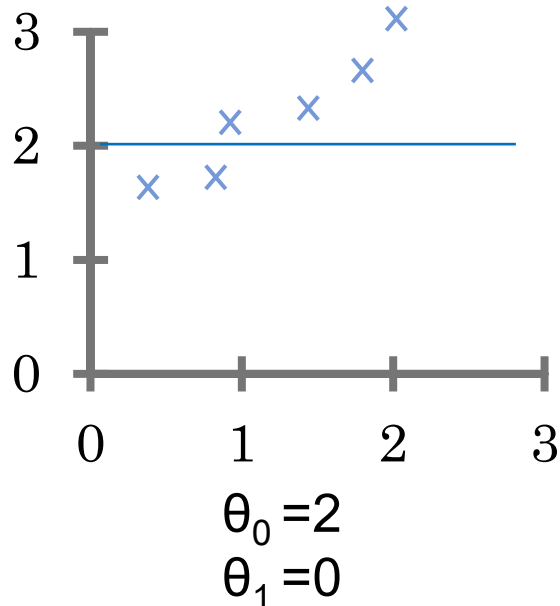
$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

univariate

2.1. UNIVARIATE LINEAR REGRESSION

- Hypothesis (h): $h_{\theta}(x) = \theta_0 + \theta_1 x \Rightarrow$ identify θ_i ?
- Method: “try_and_error”, evaluate the ability of the regression line in describing sample data.

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$



2.1. UNIVARIATE LINEAR REGRESSION

- Chose (θ_0, θ_1) so that $h_{\theta}(x^{(i)}) \approx y^{(i)}$; $i=1 \dots N$
 - residual/prediction error

$$e = h_{\theta}(x^{(i)}) - y^{(i)}$$

- MSE

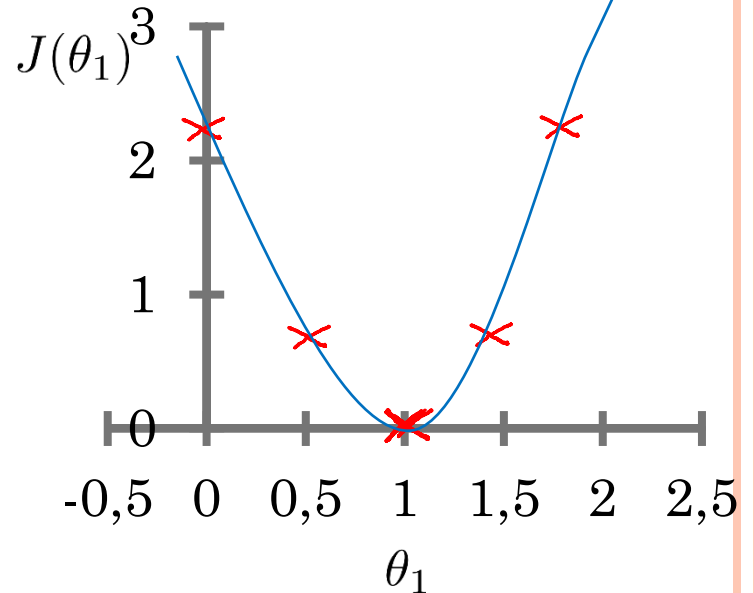
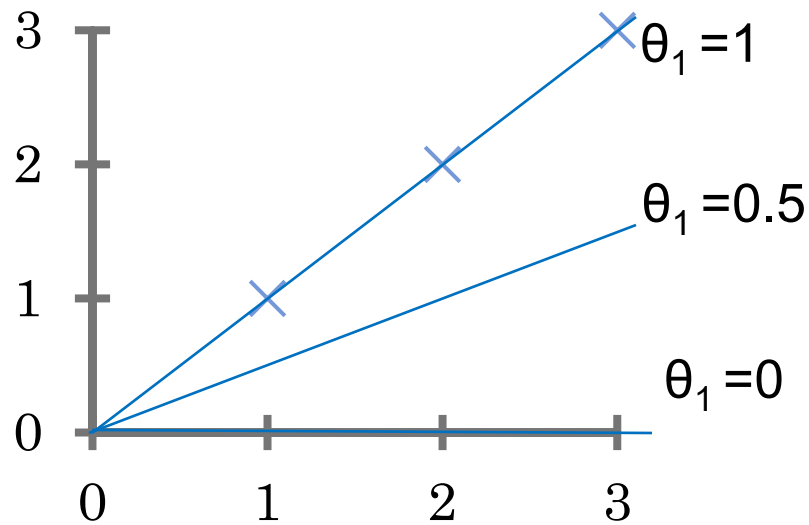
$$MSE = \frac{1}{N} \sum_{i=1}^N (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

- Cost function $J(\theta_0, \theta_1) \Rightarrow$ *minimize*

$$J(\theta_0, \theta_1) = \frac{1}{2N} \sum_{i=1}^N (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

2.1. UNIVARIATE LINEAR REGRESSION

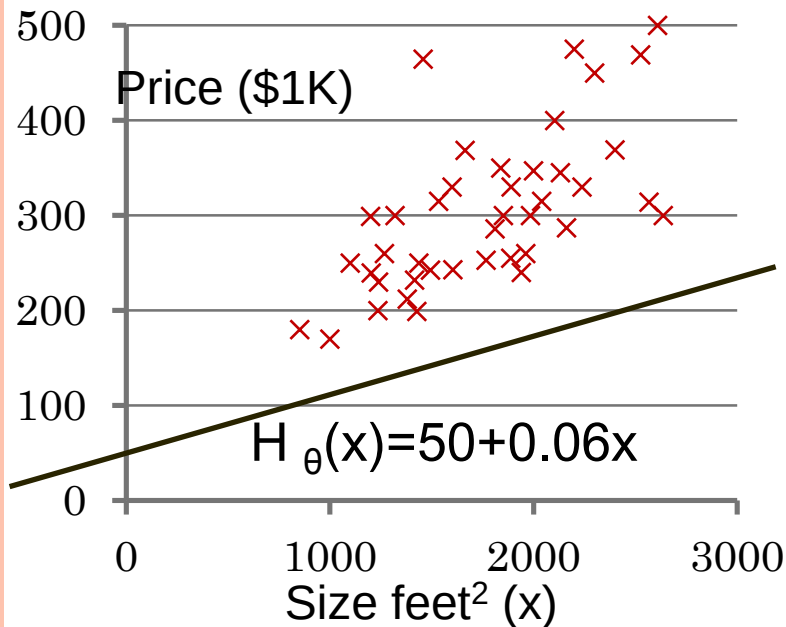
- Examine a simple case: $\theta_0=0$, $h_{\theta}(x) = \theta_1 x$



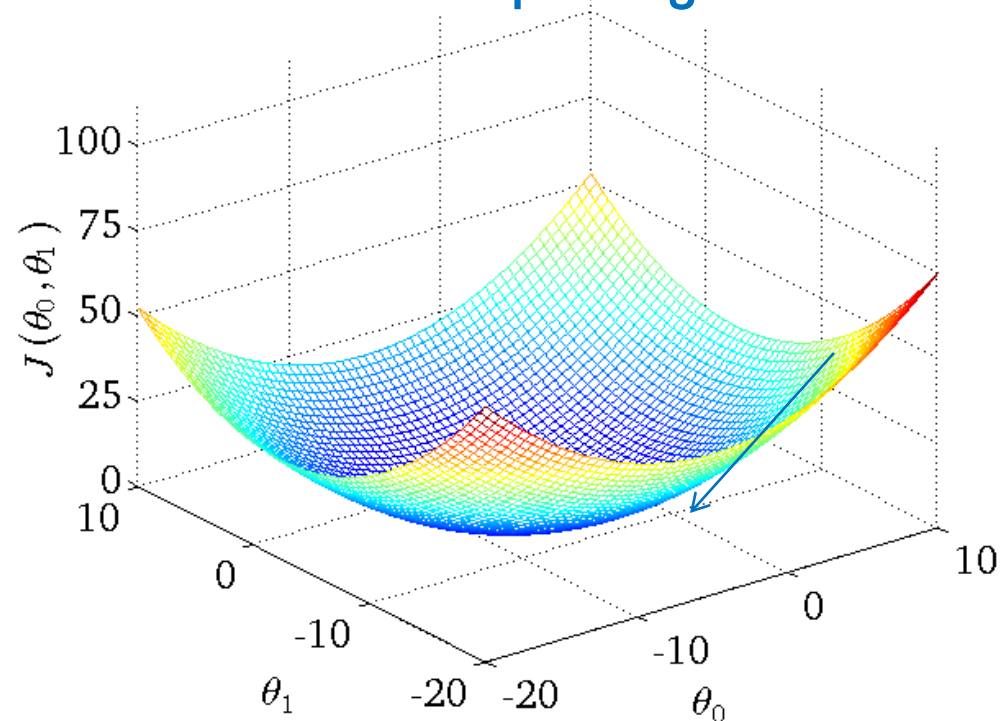
- $\theta_1 = 1 \Rightarrow J(\theta_1) = 0$; $\theta_1 = 0.5 \Rightarrow J(\theta_1) = 0.58$;
 $\theta_1 = 0 \Rightarrow J(\theta_1) = 2.3$

2.1. UNIVARIATE LINEAR REGRESSION

- An example with $h_{\theta}(x) = \theta_0 + \theta_1 x$



Contour plots/figures

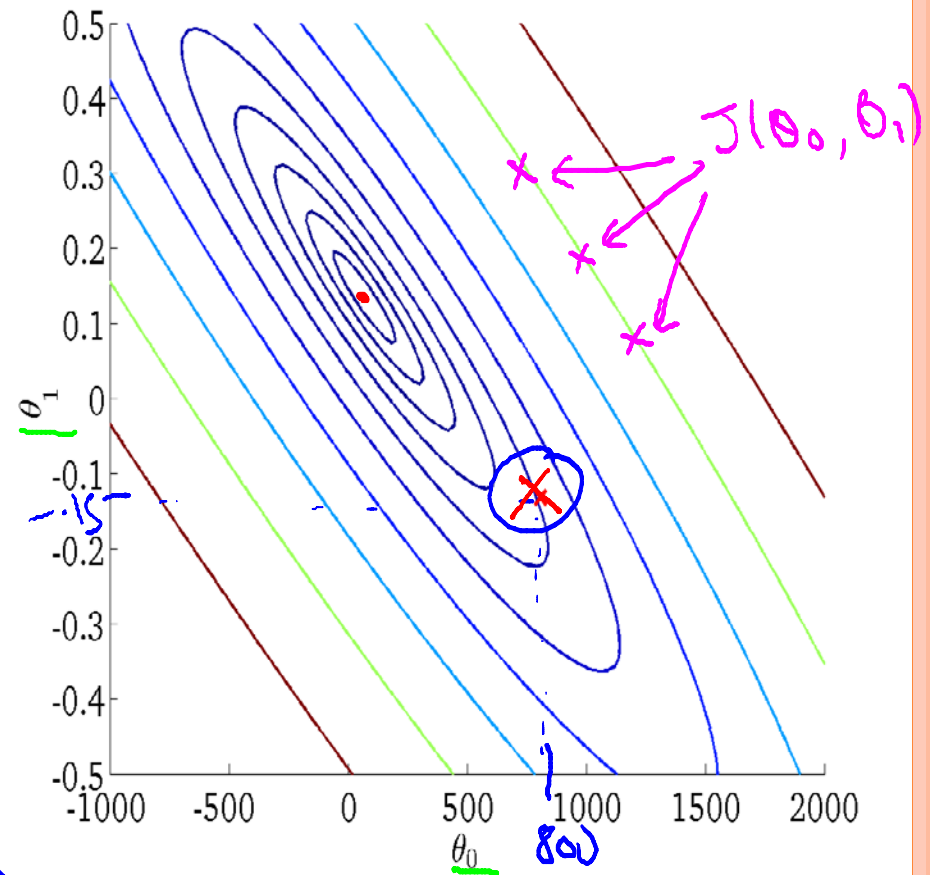
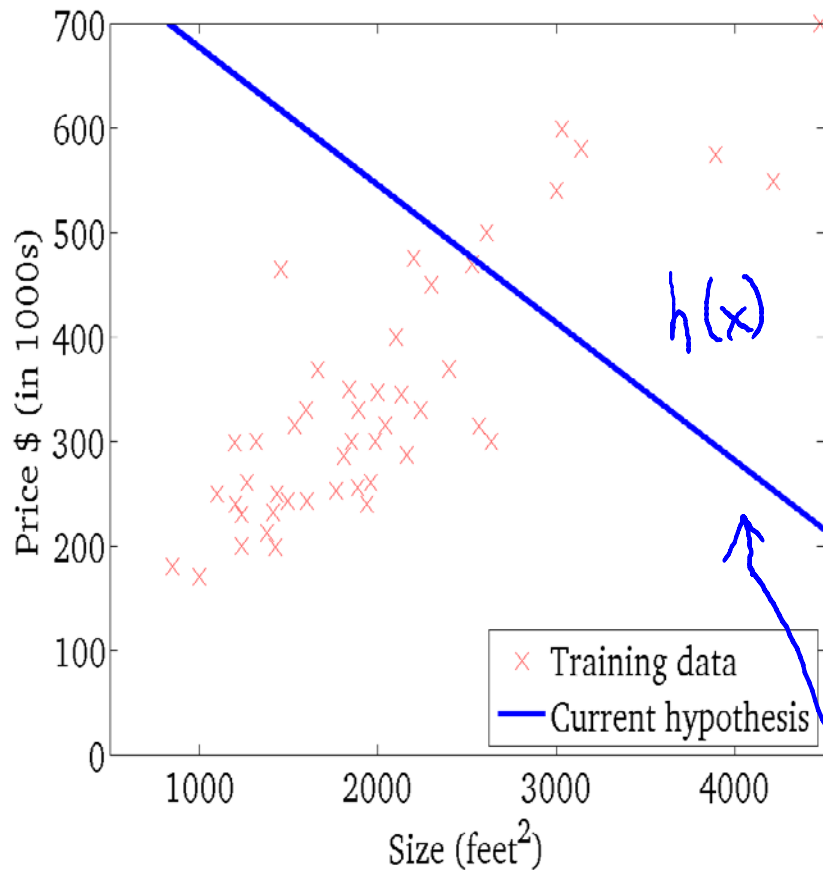


$$h_{\theta}(x)$$

$$J(\theta_0, \theta_1)$$

(for fixed θ_0, θ_1 , this is a function of x)

(function of the parameters θ_0, θ_1)

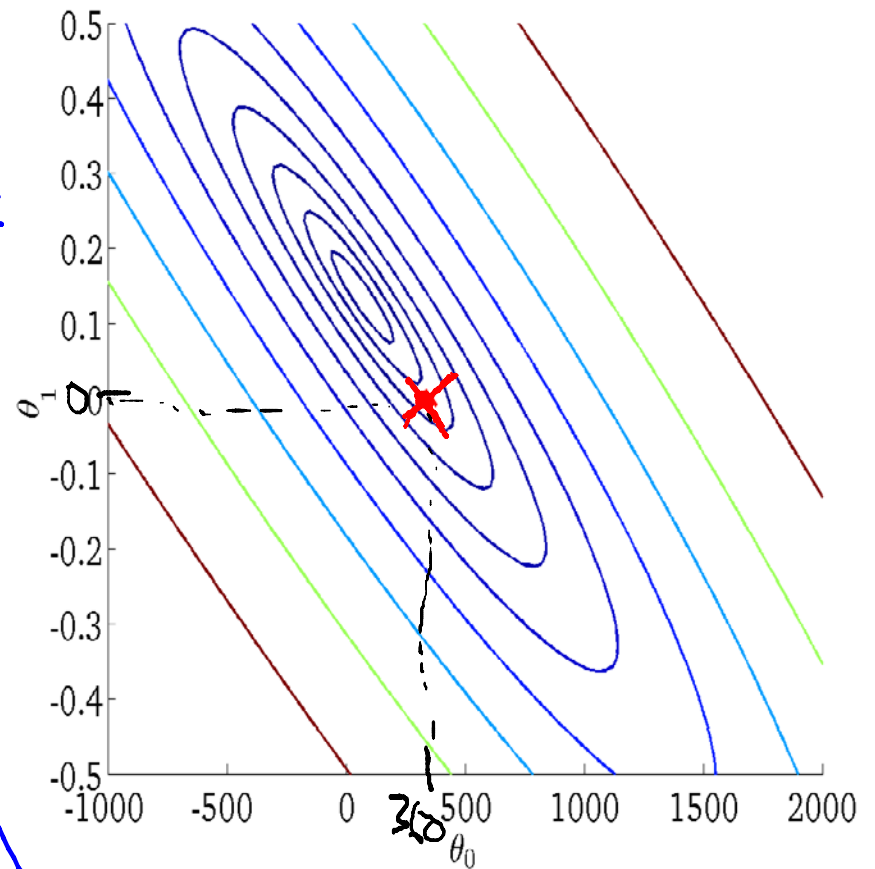
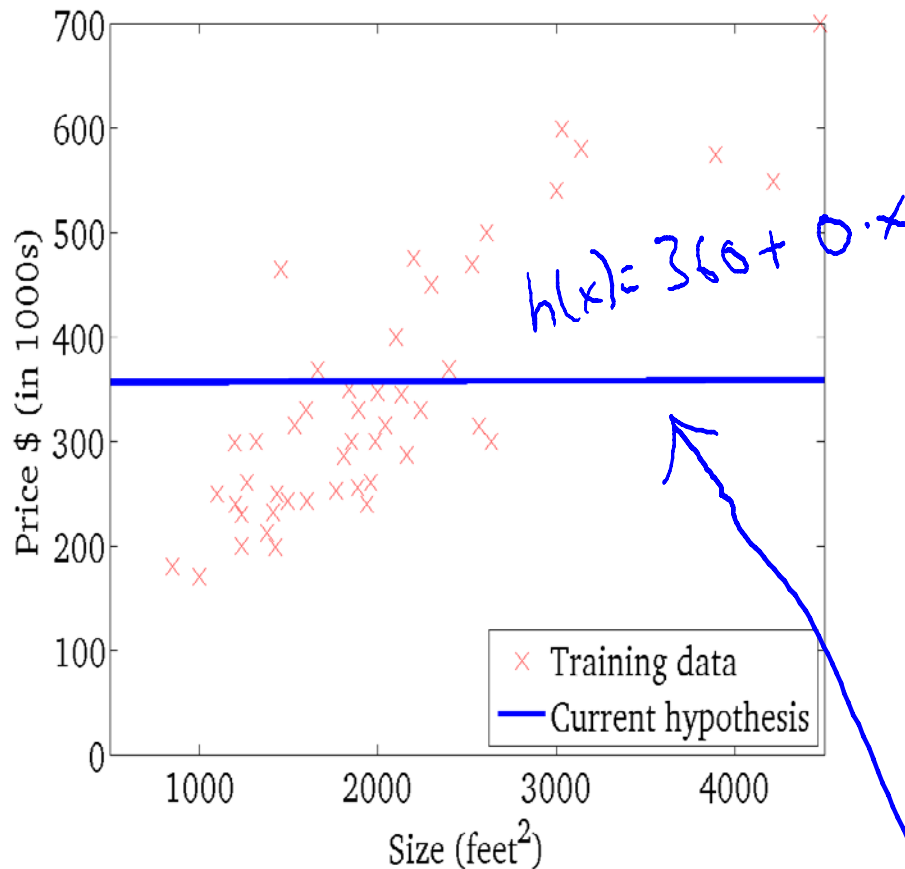


$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)

$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)

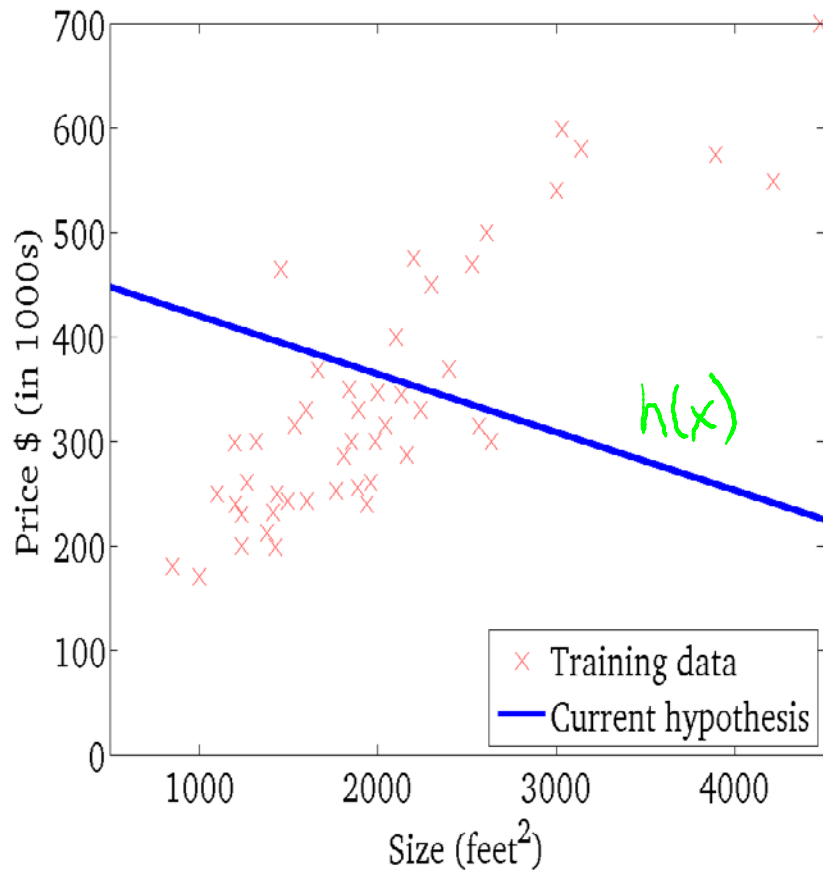


$$\begin{cases} \theta_0 = 360 \\ \theta_1 = 0 \end{cases}$$



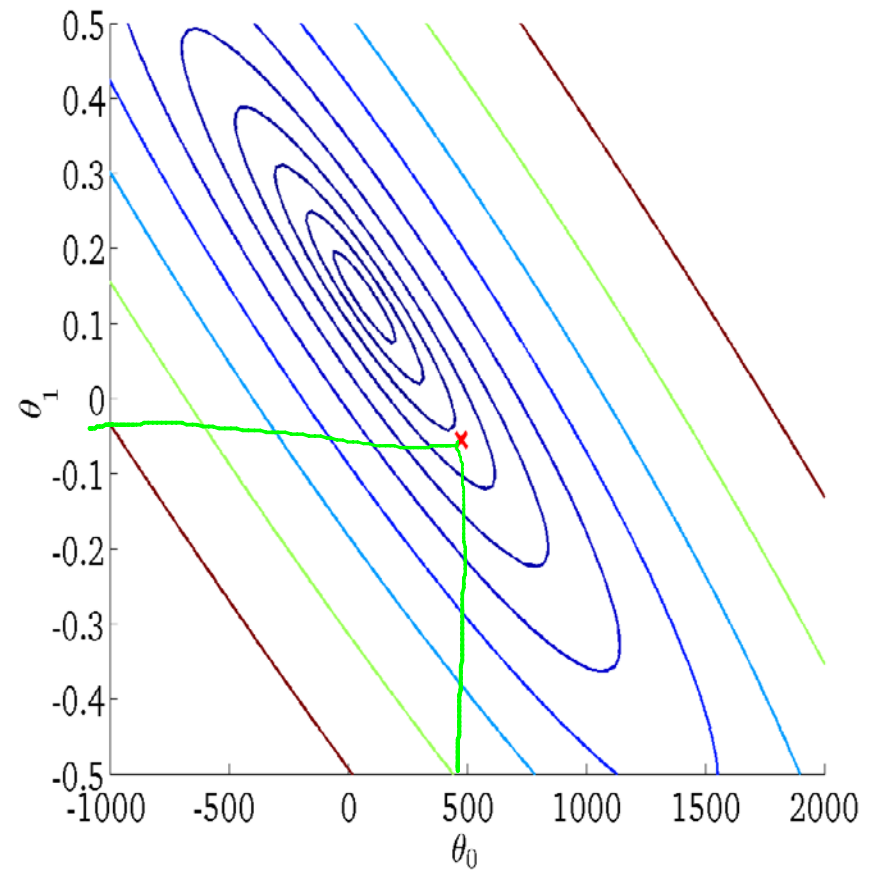
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



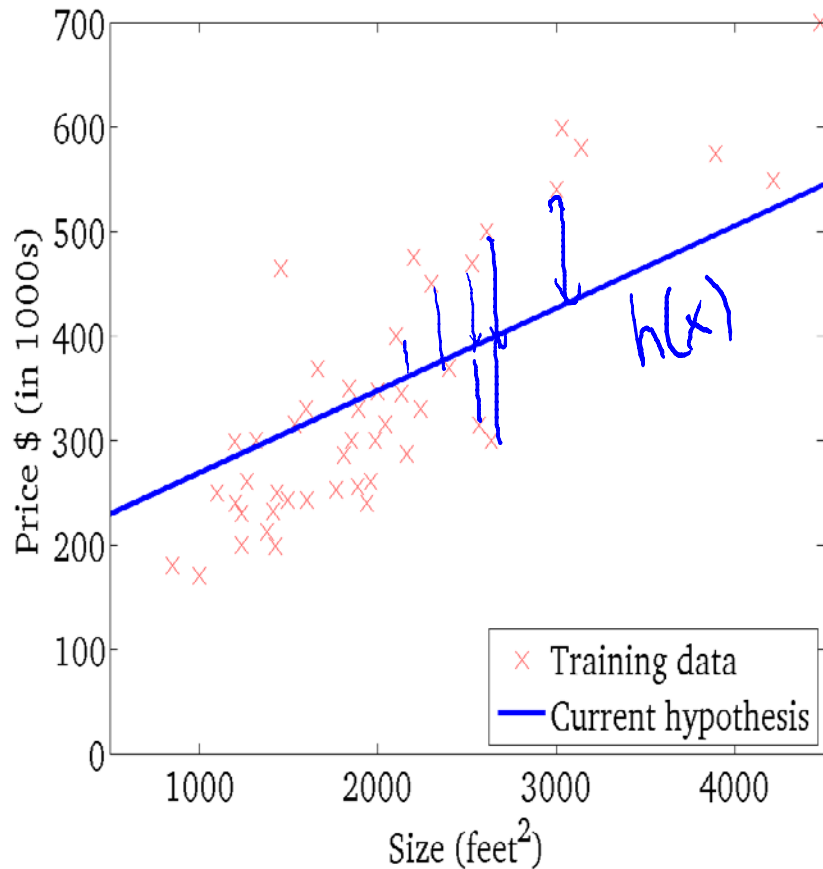
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



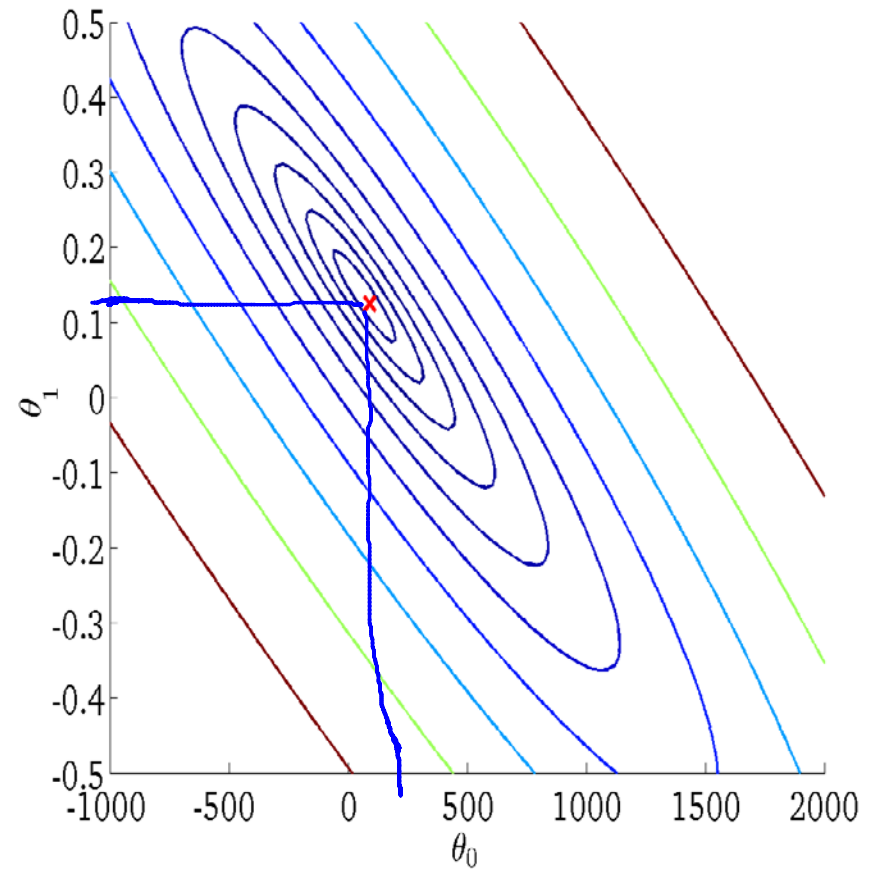
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



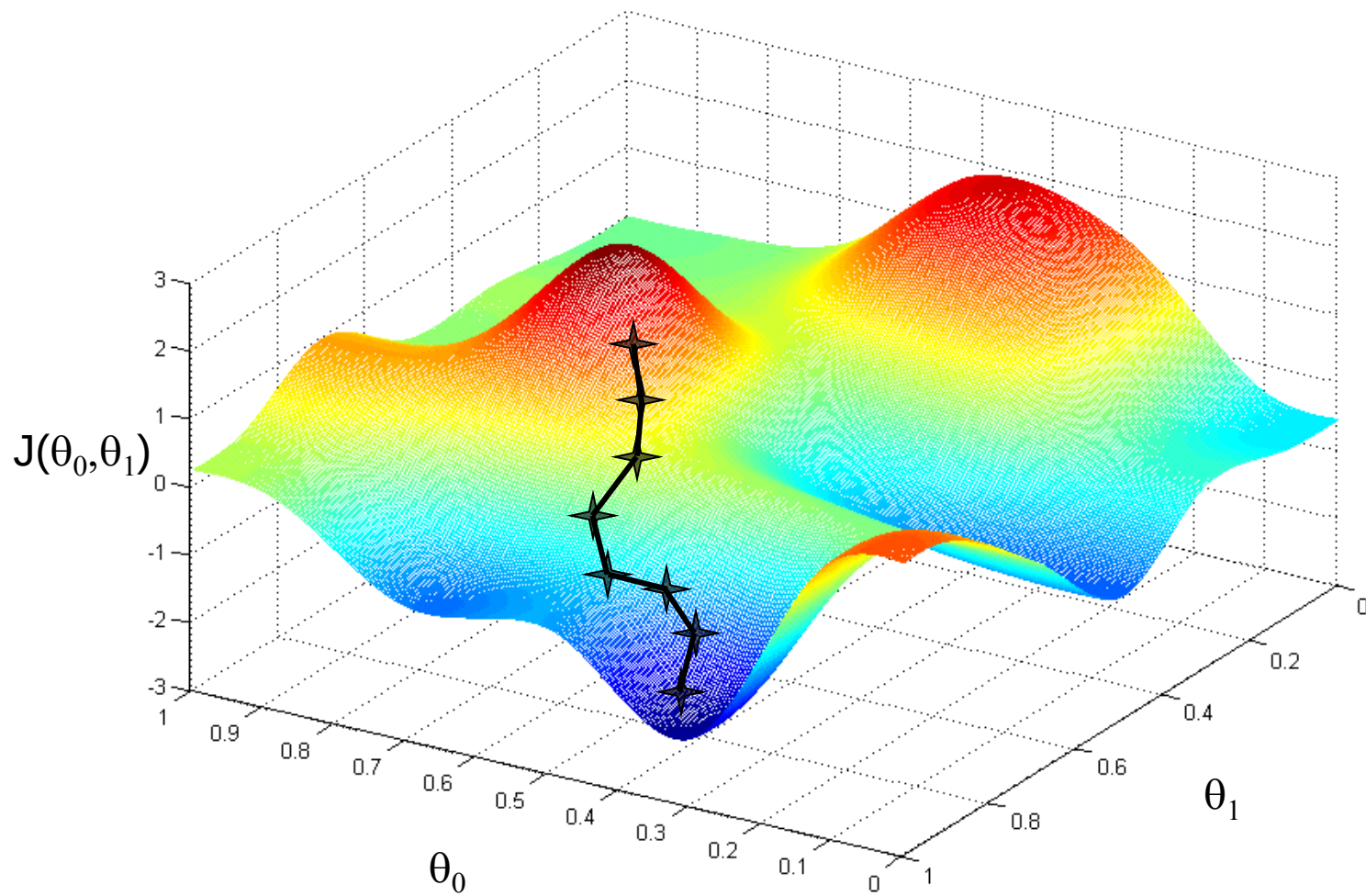
$$J(\theta_0, \theta_1)$$

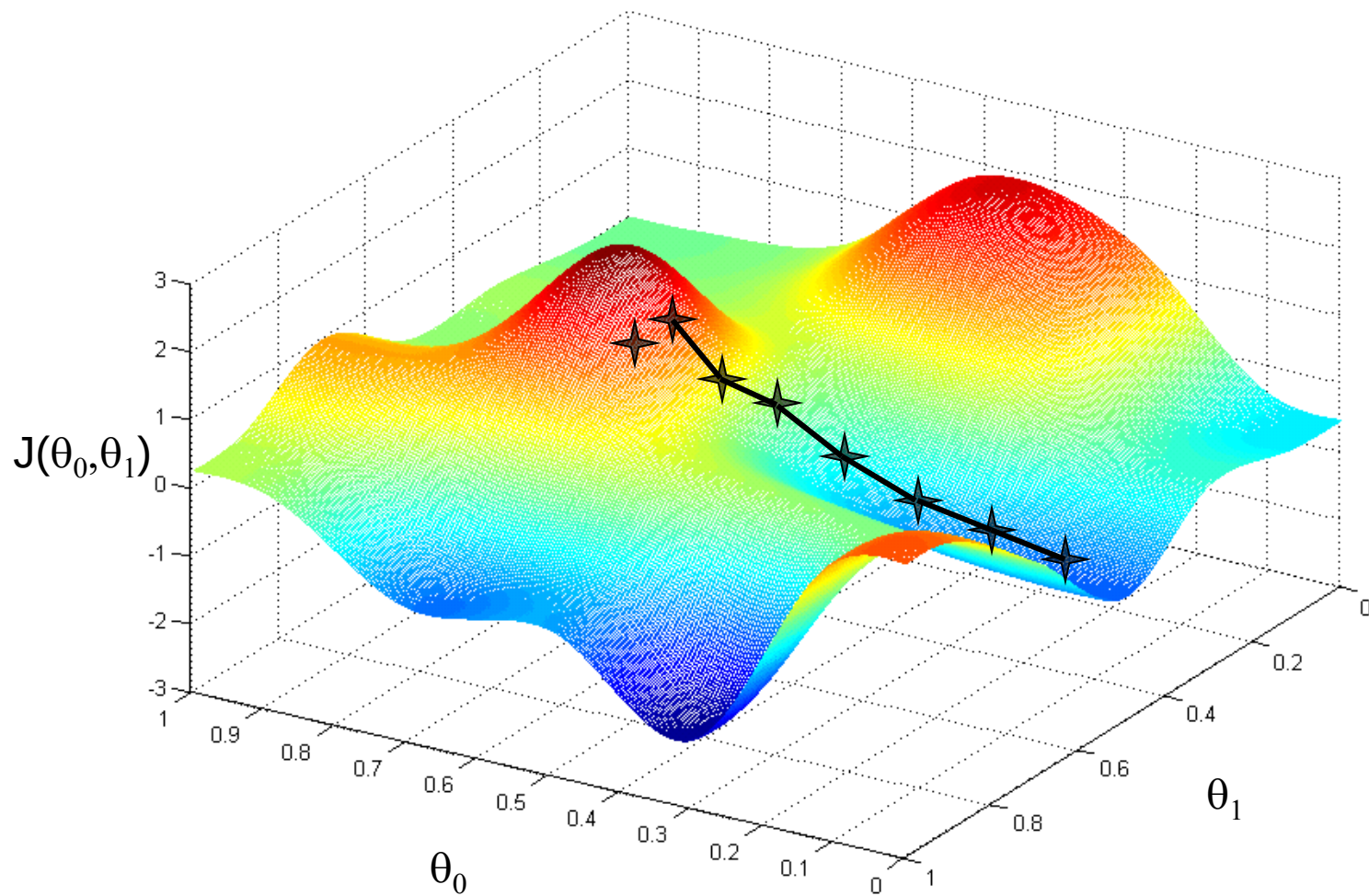
(function of the parameters θ_0, θ_1)



2.1. UNIVARIATE LINEAR REGRESSION

- Gradient descent method $\Rightarrow$ find our the point that minimize $J(\theta_0, \theta_1)$
- Method:
 - i. Initiate with a random parameter (θ_0, θ_1) , ex. $(\theta_0=0, \theta_1=0)$
 - ii. Change (θ_0, θ_1) to reduce $J(\theta_0, \theta_1)$
 - iii. Iterate step ii until $J(\theta_0, \theta_1)$ is (or we believe/accept that it is) minimum





2.1. UNIVARIATE LINEAR REGRESSION

○ Gradient descent algorithm

Repeat until convergence{

$$\theta_j = \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) \quad // \text{ for } j=0 \text{ and } j=1, \text{ simultaneously}$$

}

Learning rate

Correct : Simultaneously update

$$\text{temp0} := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\text{temp1} := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$\theta_0 := \text{temp0}$$

$$\theta_1 := \text{temp1}$$

Wrong:

$$\text{temp0} := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\theta_0 := \text{temp0}$$

$$\text{temp1} := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$\theta_1 := \text{temp1}$$

2.1. UNIVARIATE LINEAR REGRESSION

- Gradient descent algorithm: minimize $J(\theta_0, \theta_1)$

$$J(\theta_0, \theta_1) = \frac{1}{2N} \sum_{i=1}^N (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Repeat until convergence{

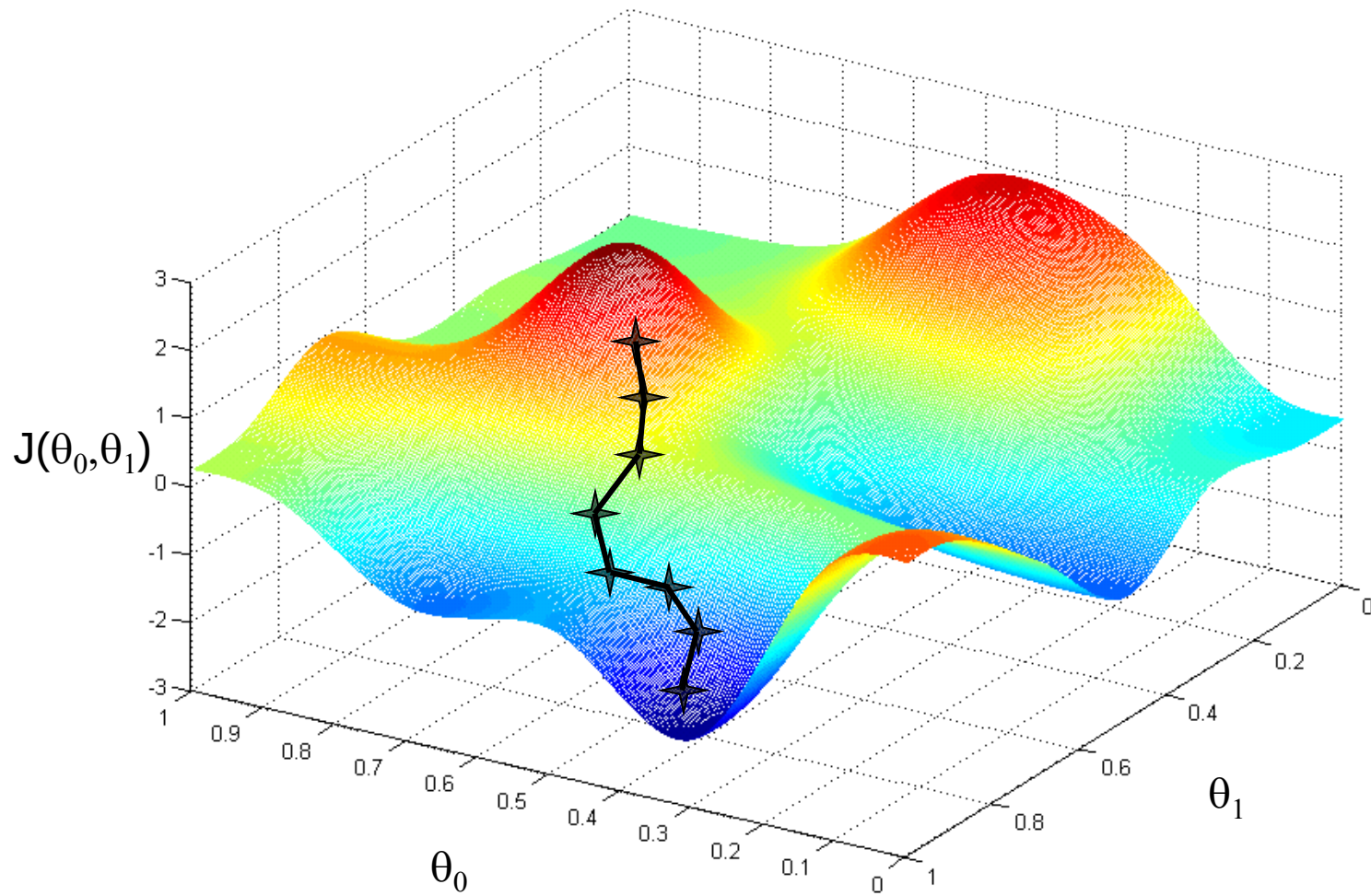
$$\theta_0 = \theta_0 - \alpha \left[\frac{1}{N} \sum_{i=1}^N (h_{\theta}(x^{(i)}) - y^{(i)}) \right]$$

$$\theta_1 = \theta_1 - \alpha \left[\frac{1}{N} \sum_{i=1}^N (h_{\theta}(x^{(i)}) - y^{(i)}) x^{(i)} \right]$$

}

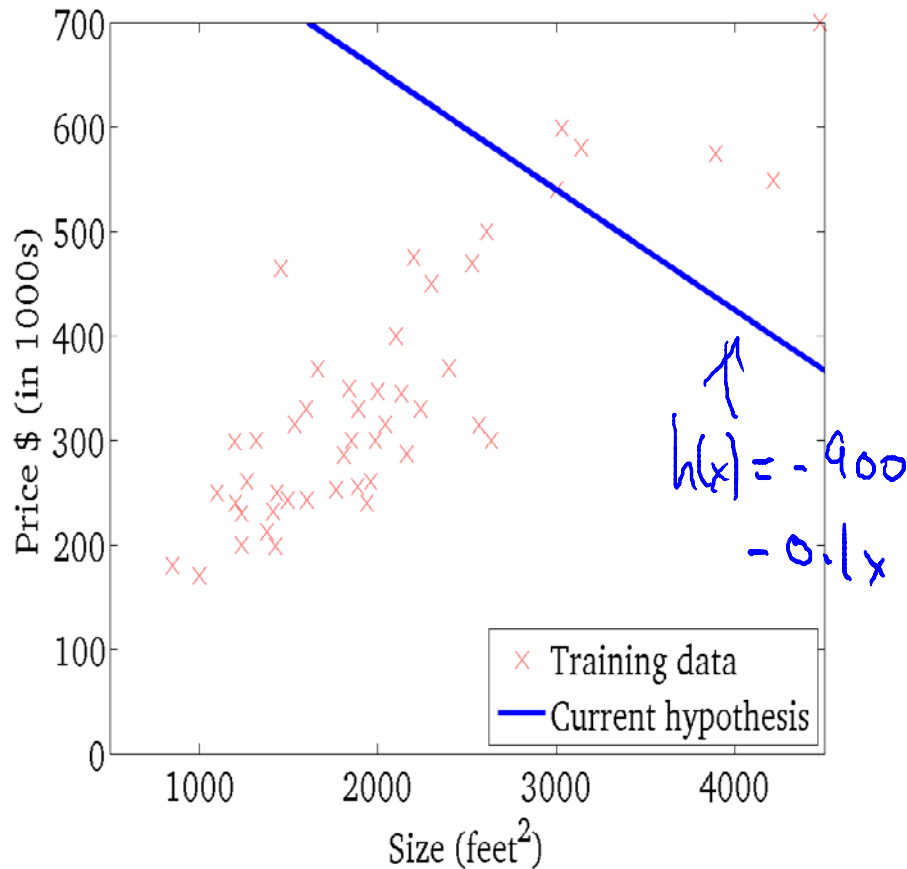
Update θ_0 and θ_1 simultaneously

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$



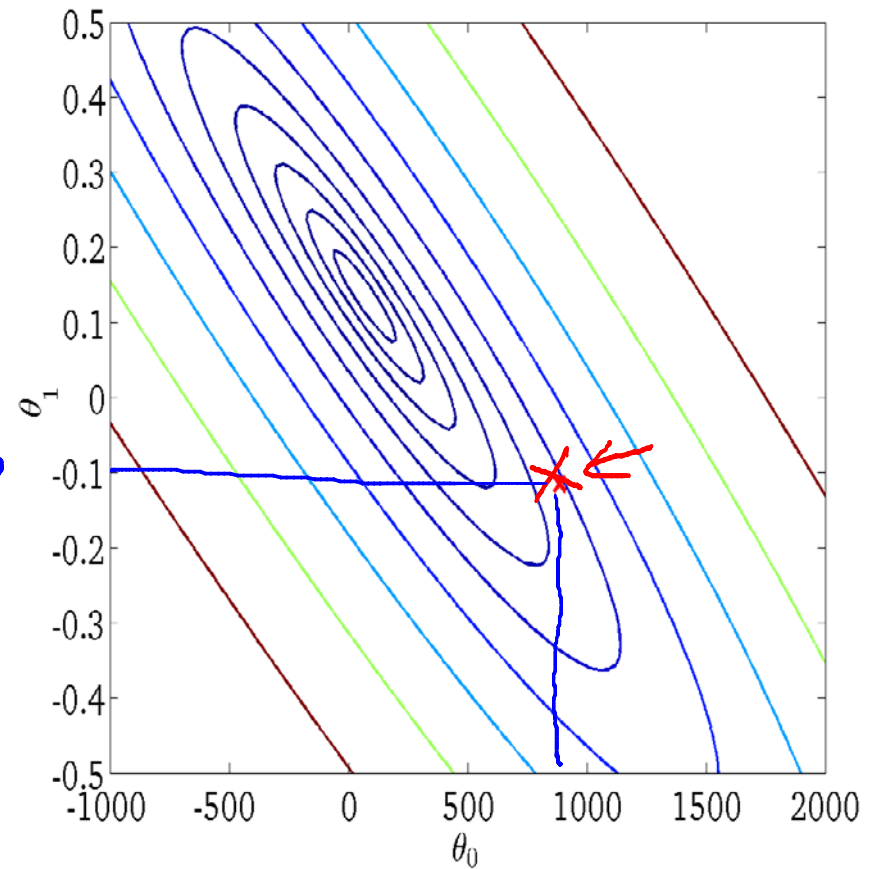
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



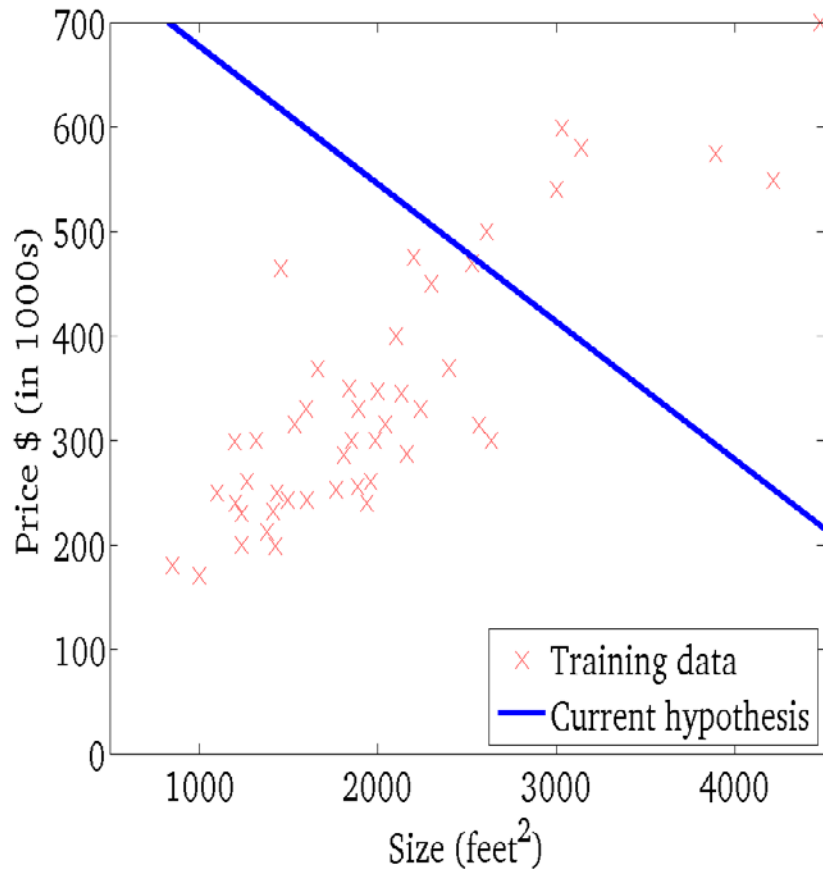
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



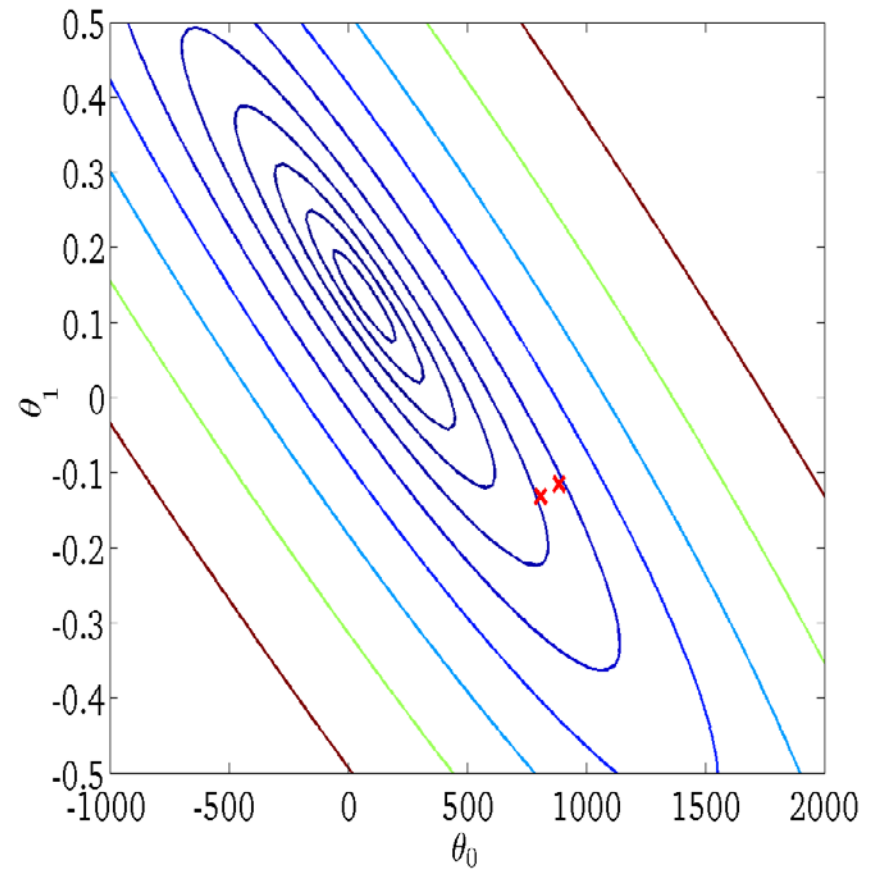
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(for fixed θ_0, θ_1 , this is a function of x)



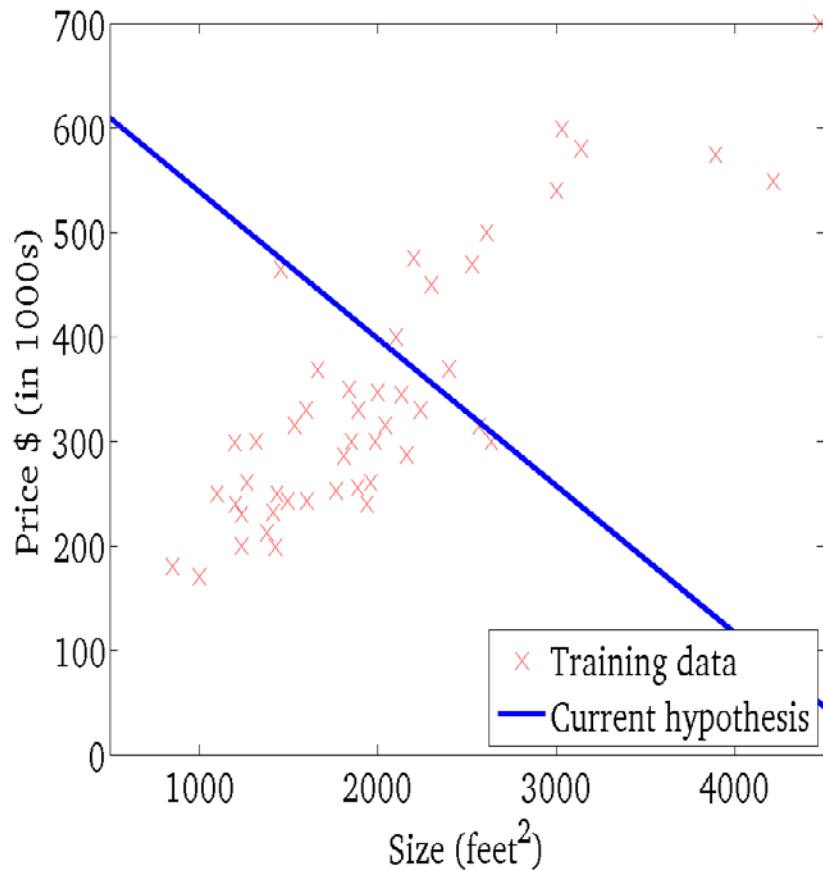
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



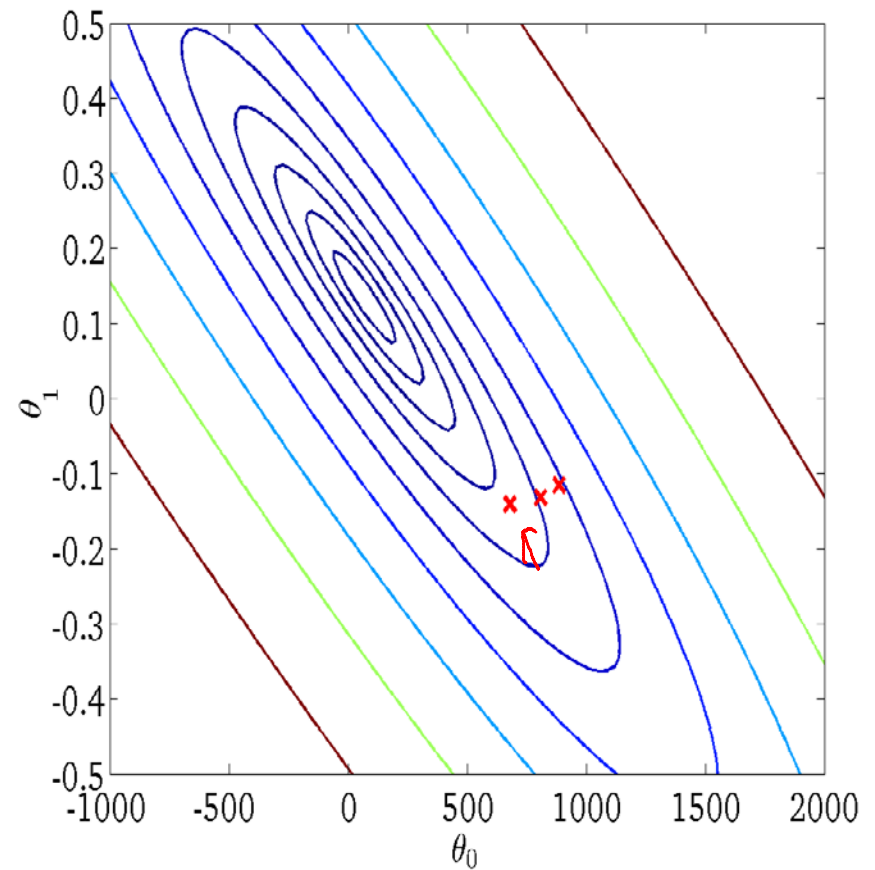
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(for fixed θ_0, θ_1 , this is a function of x)



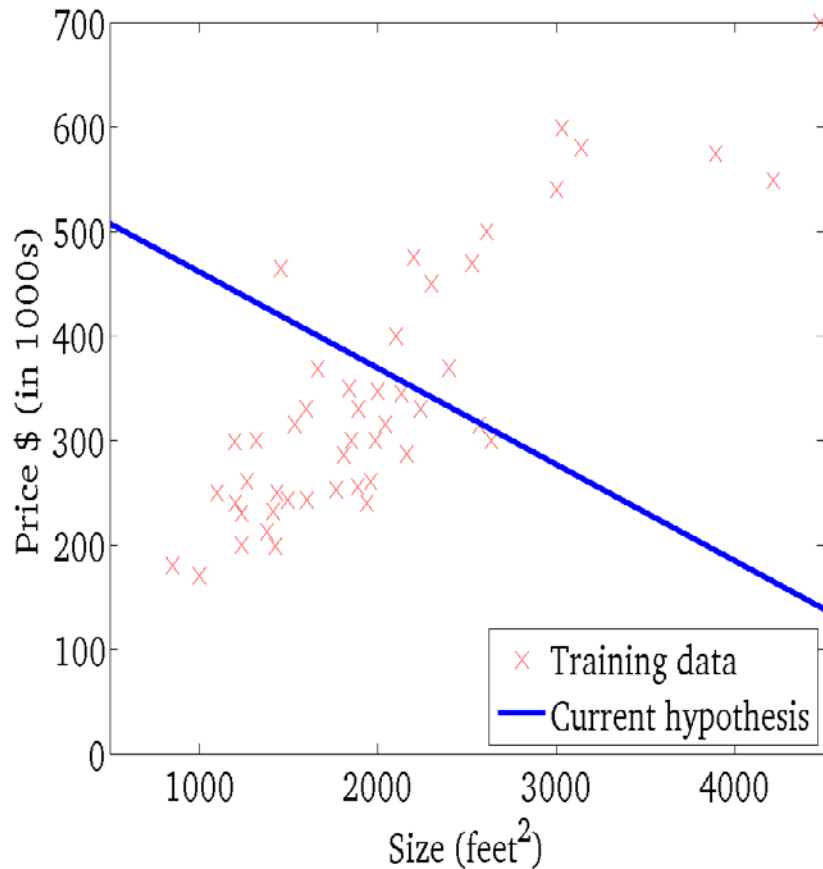
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



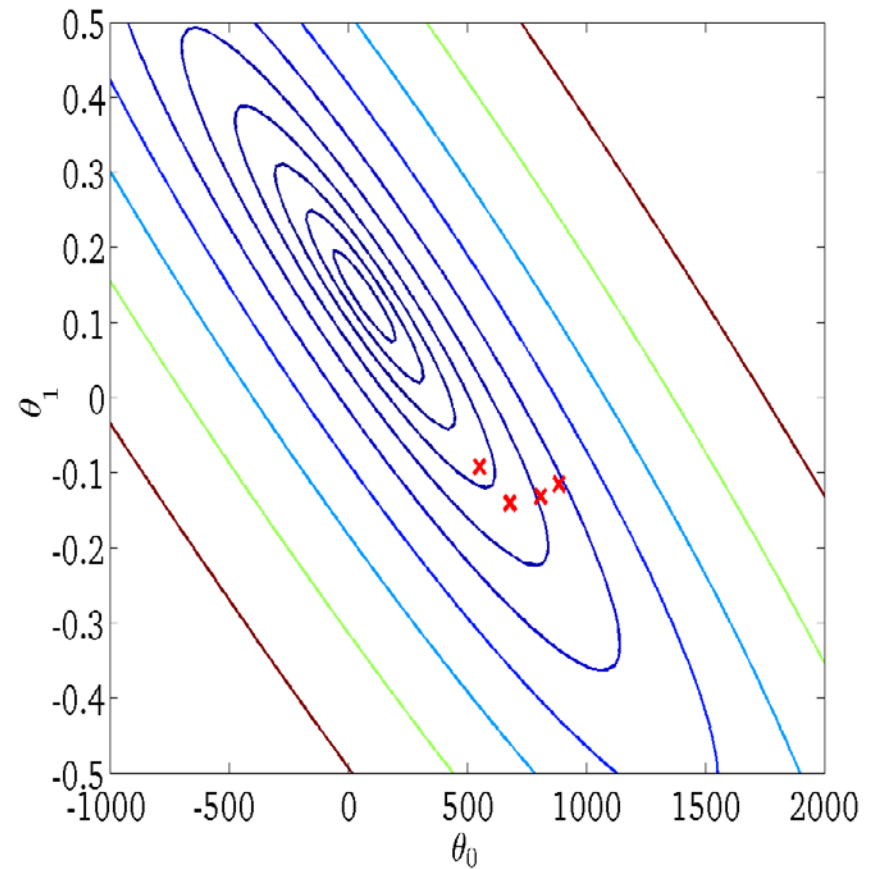
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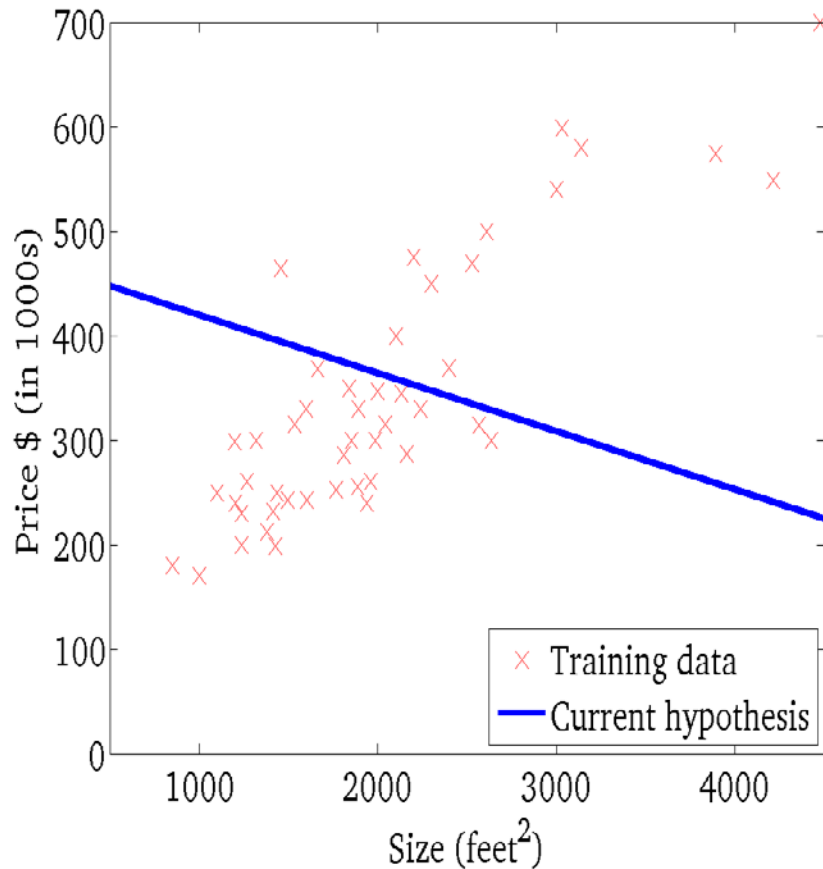
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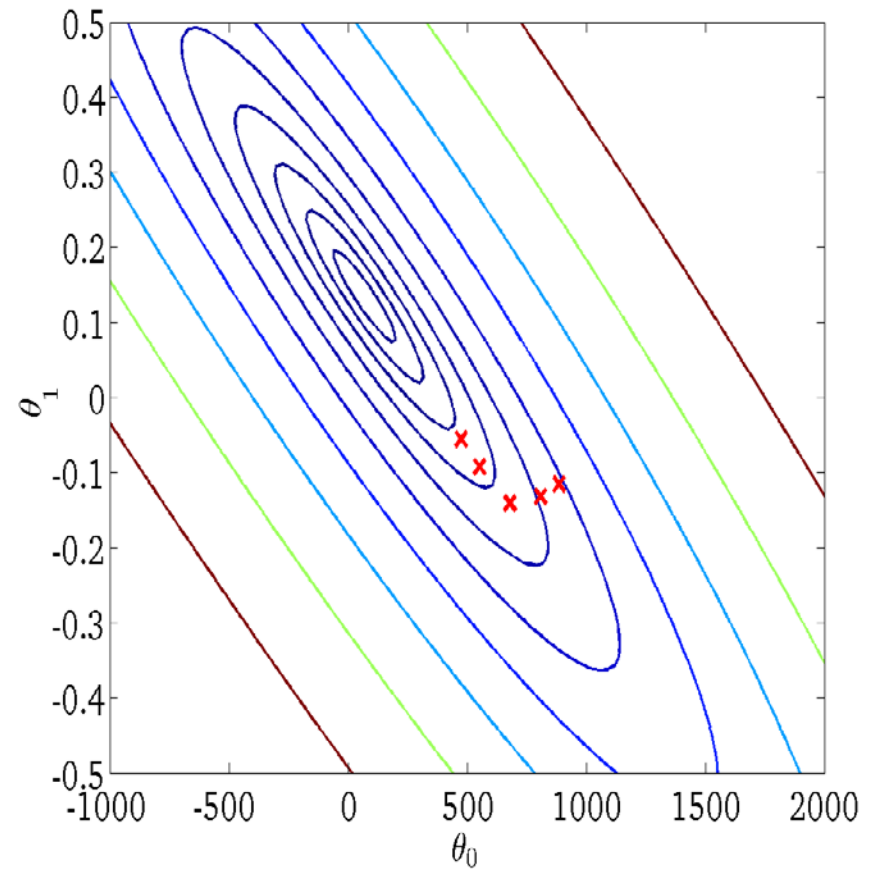
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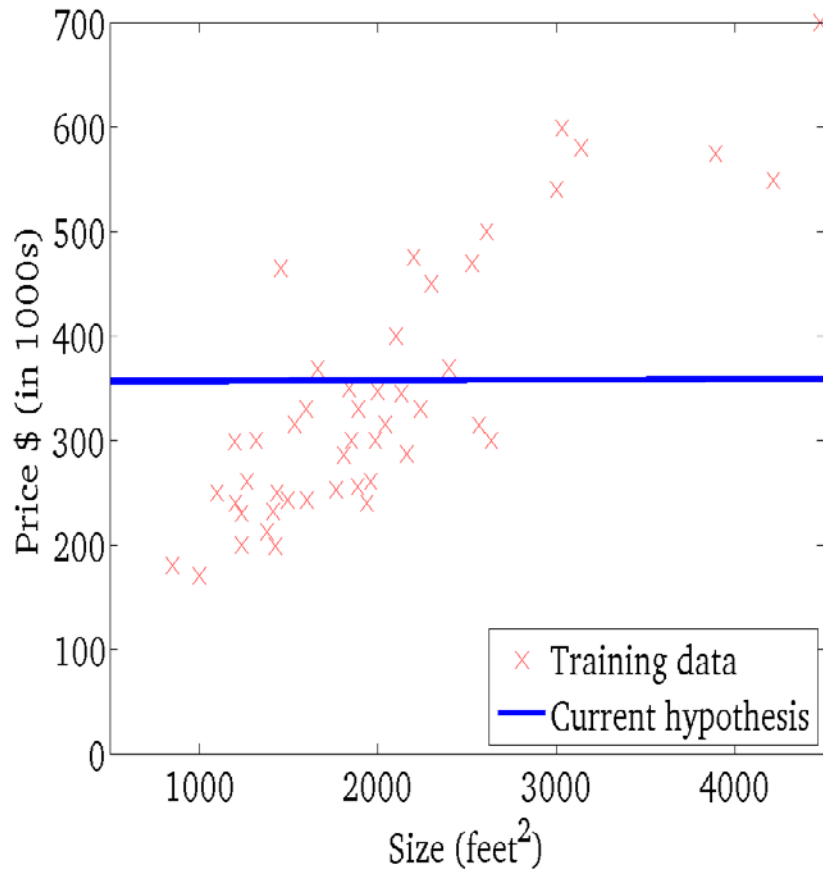
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



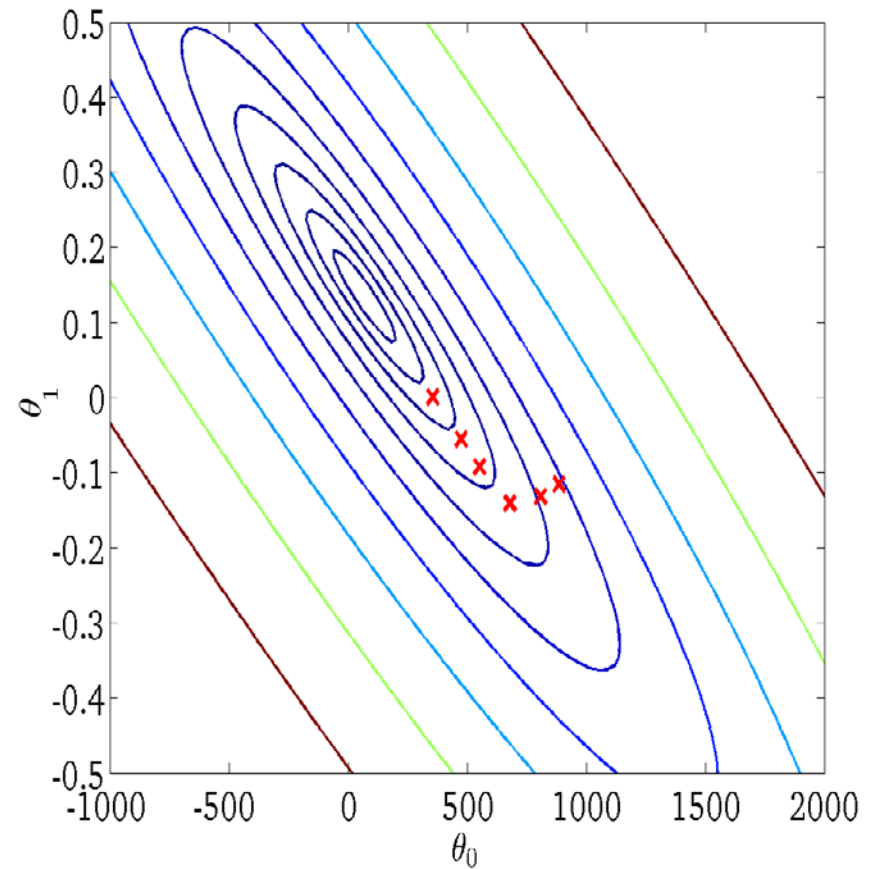
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



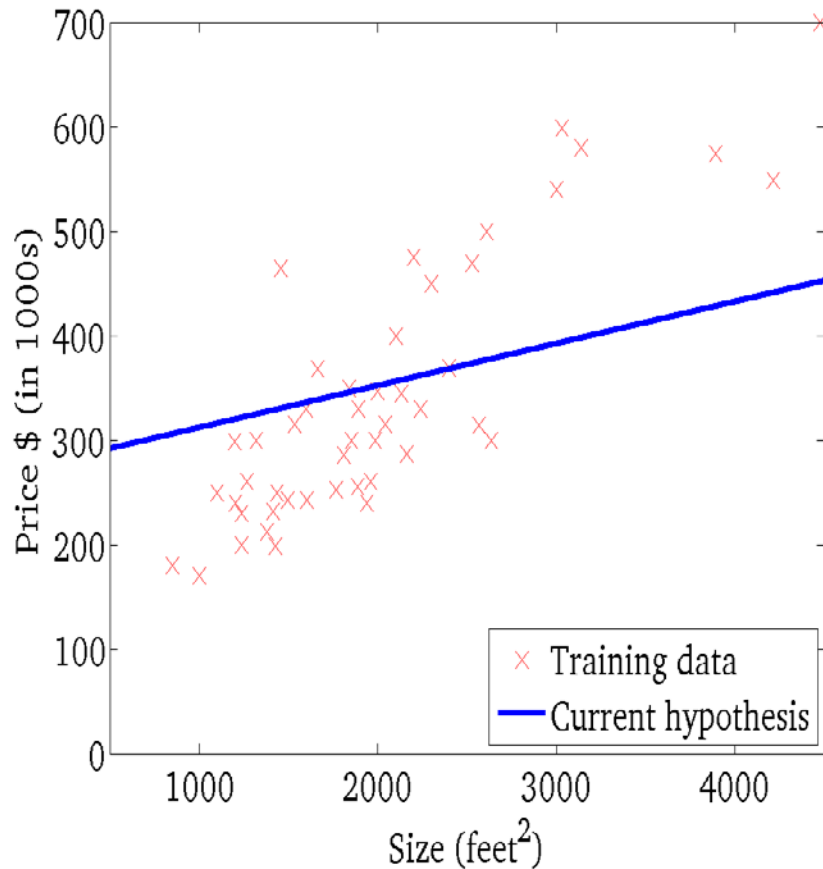
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



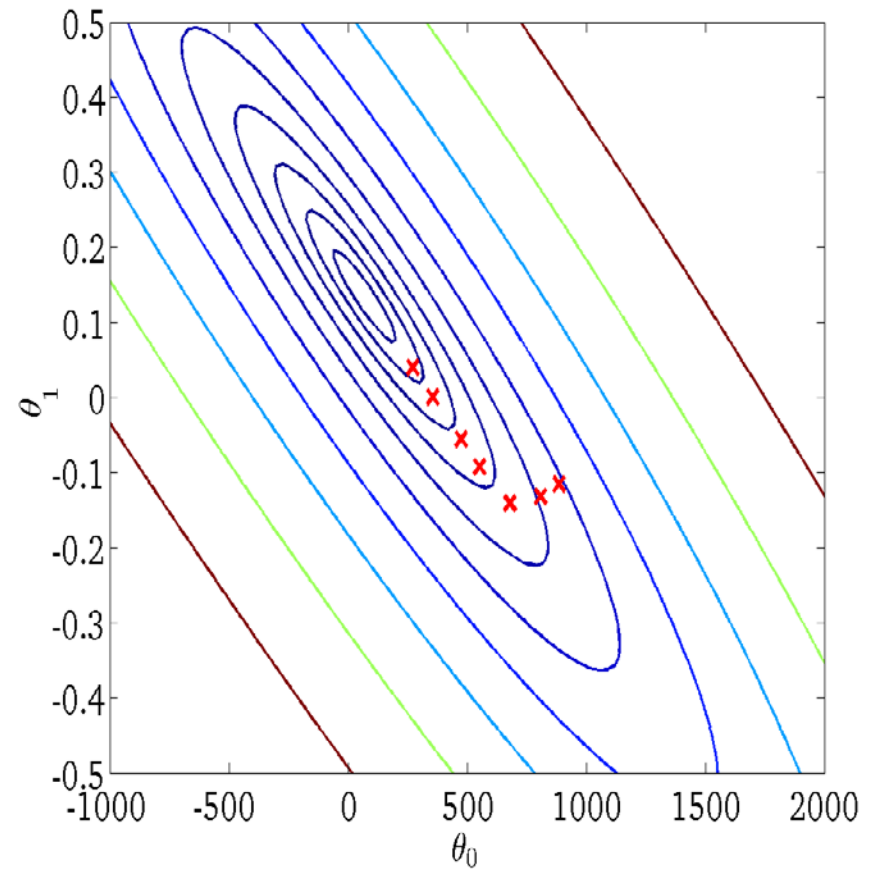
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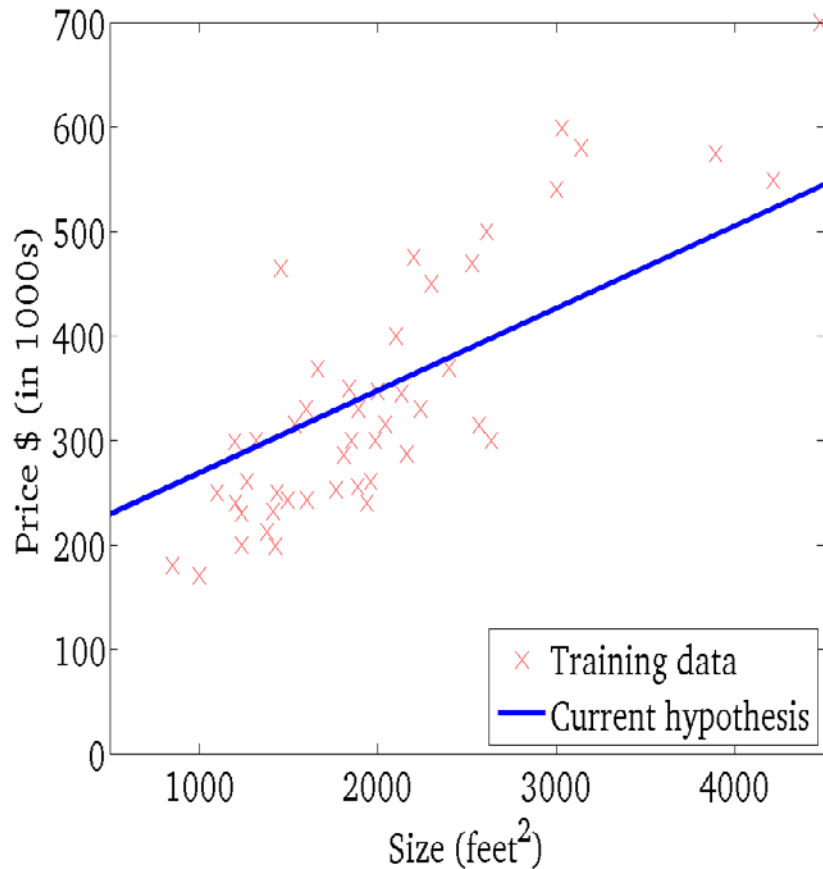
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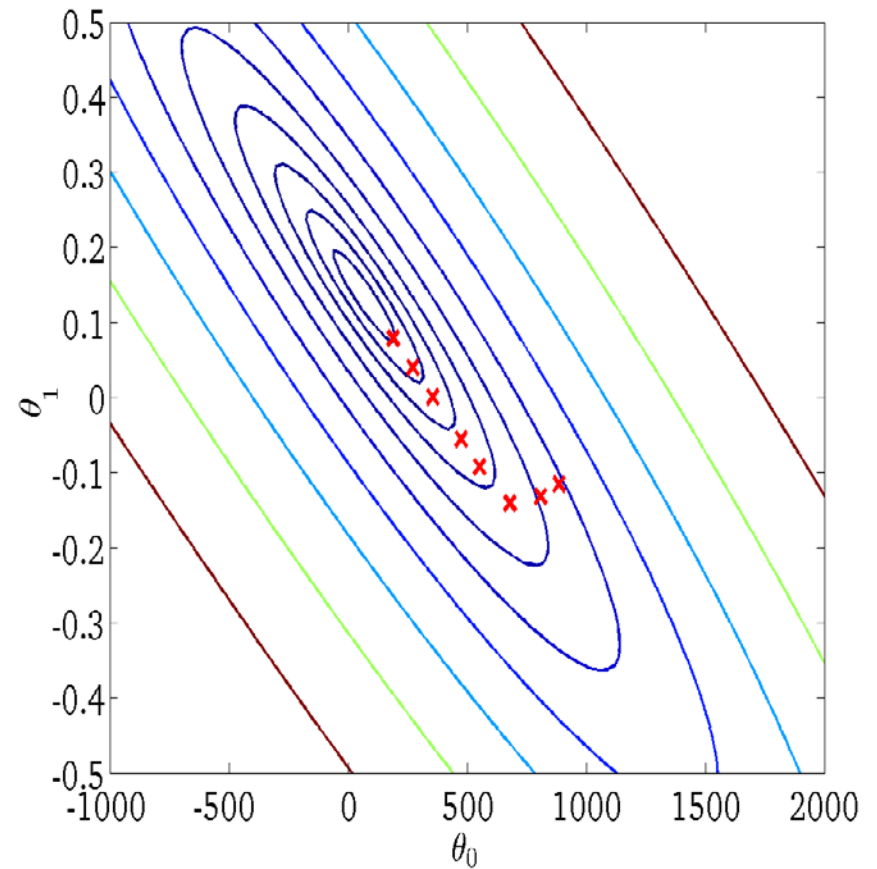
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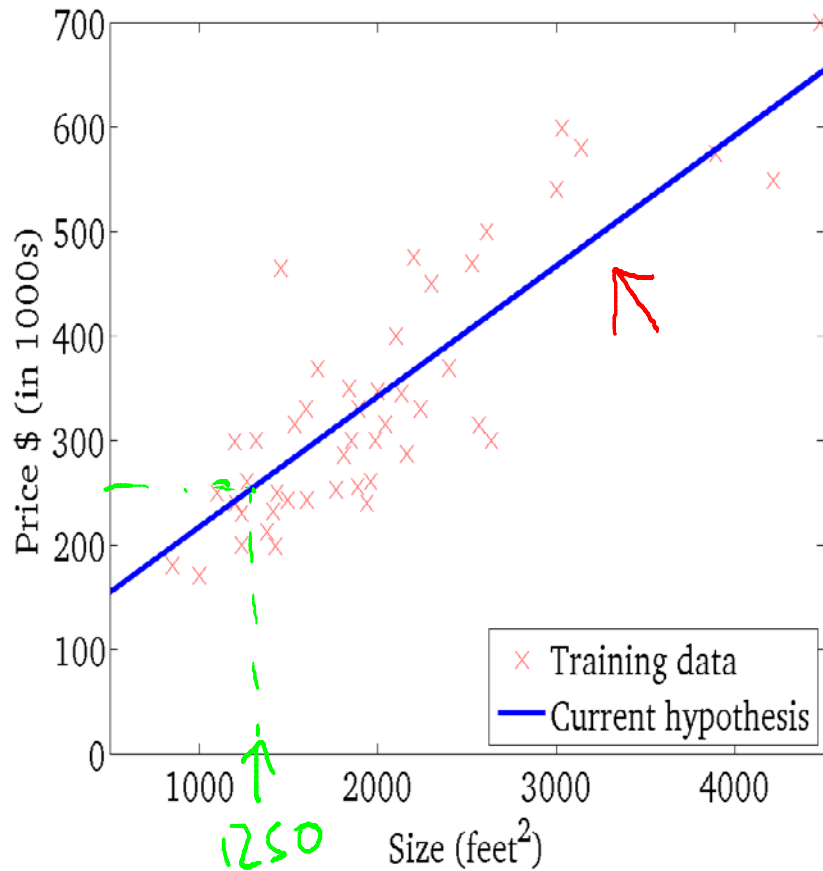
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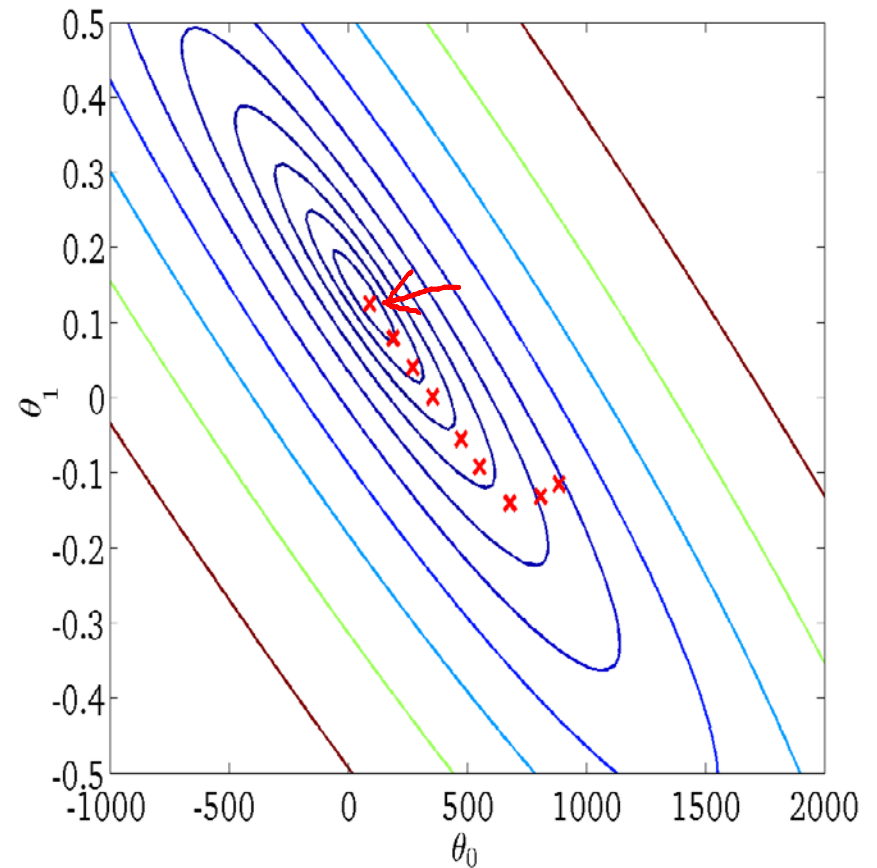
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(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)

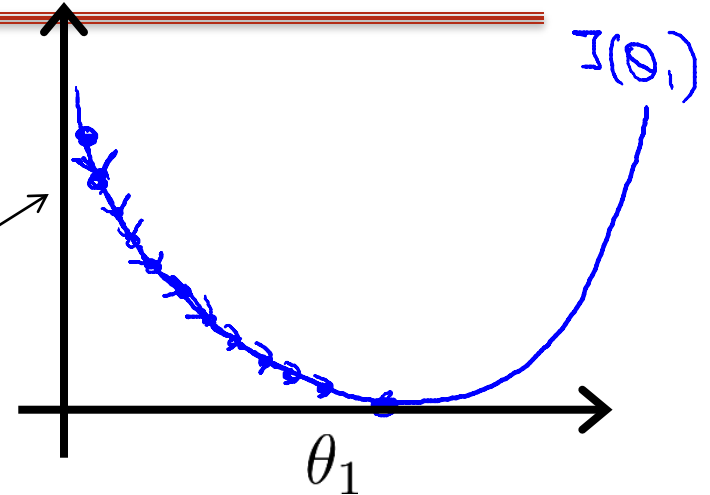


2.1. UNIVARIATE LINEAR REGRESSION

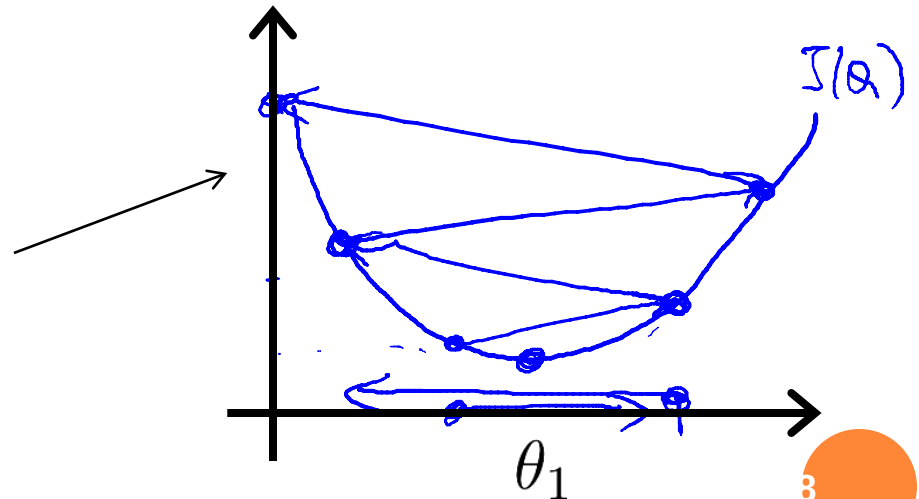
- Learning rate α

$$\theta_1 = \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_1)$$

- ✓ Too small: Learn slowly



- ✓ Too big: Difficult to converge to the point where $J(\theta_0, \theta_1)$ is minimum



2.1. UNIVARIATE LINEAR REGRESSION

- Summary: Given N samples, a univariate regression model is described as follow (e_i describes the change of Y which is not explainable from X)

- ✓ Straight line form

$$y_i = h_{\theta}(x_i) = \theta_0 + \theta_1 x_i + e_i, i = 1, 2, \dots, N$$

- ✓ **Parabol** form

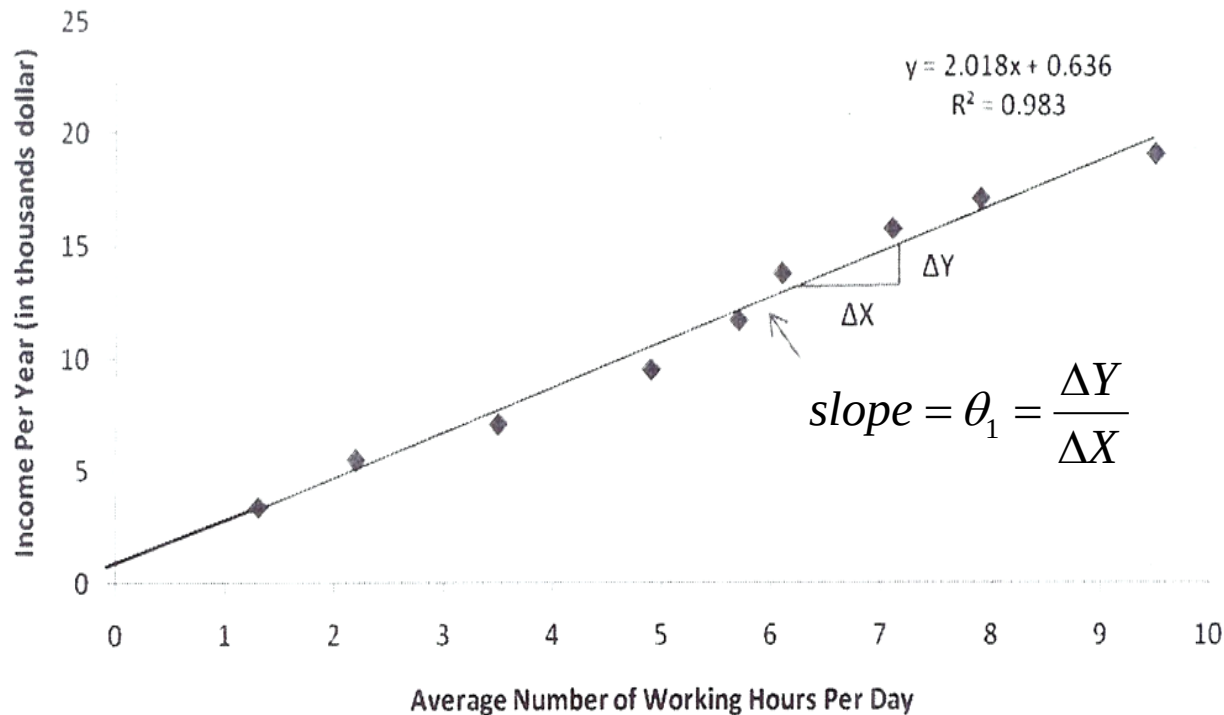
$$y_i = h_{\theta}(x_i) = \theta_0 + \theta_1 x_i + \theta_2 x_i^2 + e_i, i = 1, 2, \dots, N$$

- Estimating (θ_0, θ_1) by *gradient descent* method or can be quickly estimated by:

$$\theta_1 = \frac{\sum_{i=1}^N (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})}{\sum_{i=1}^N (x^{(i)} - \bar{x})^2} \quad \theta_0 = \bar{y} - \theta_1 \bar{x}$$

2.1. UNIVARIATE LINEAR REGRESSION

Income vs Average Working Hours



- $Y = \theta_0 + \theta_1 * X_1 \rightarrow Y = 0.636 + 2.018 * X$
- The sign of θ_1 describes the effect direction (positive/negative) of X on Y.

2.1. UNIVARIATE LINEAR REGRESSION

Quantity Sold	Price(\$)
8500	2
4700	5
5800	3
7400	2
6200	5
7300	3
5600	4



$$y = \text{quantitySold} = 9323 - 823 * \text{price}$$

2.1. UNIVARIATE LINEAR REGRESSION

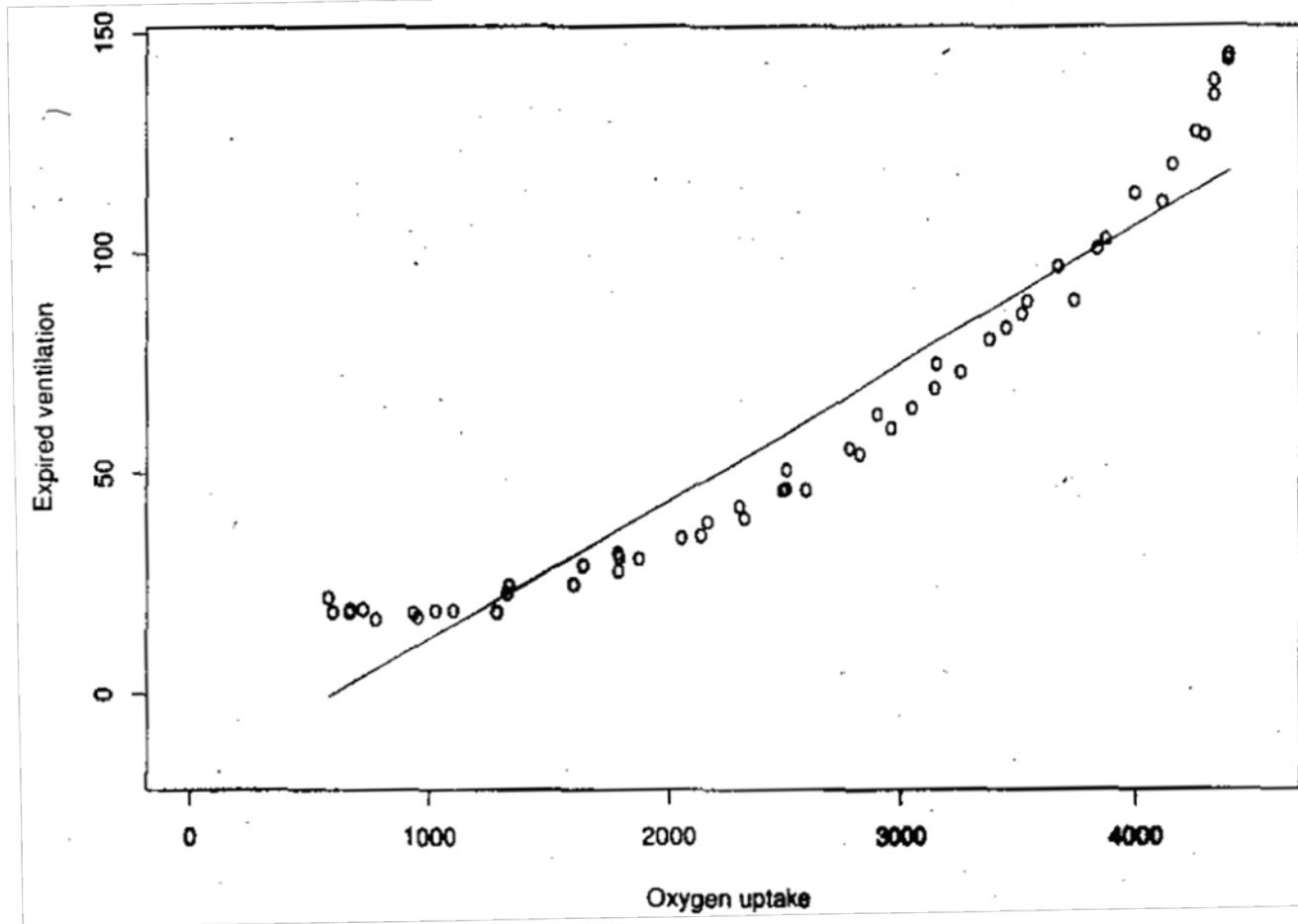


Figure 11.1, [2], pp. 372. Expired ventilation plotted against oxygen uptake in a series of trials, with fitted straight line: $y = \theta_0 + \theta_1 x$.

2.1. UNIVARIATE LINEAR REGRESSION

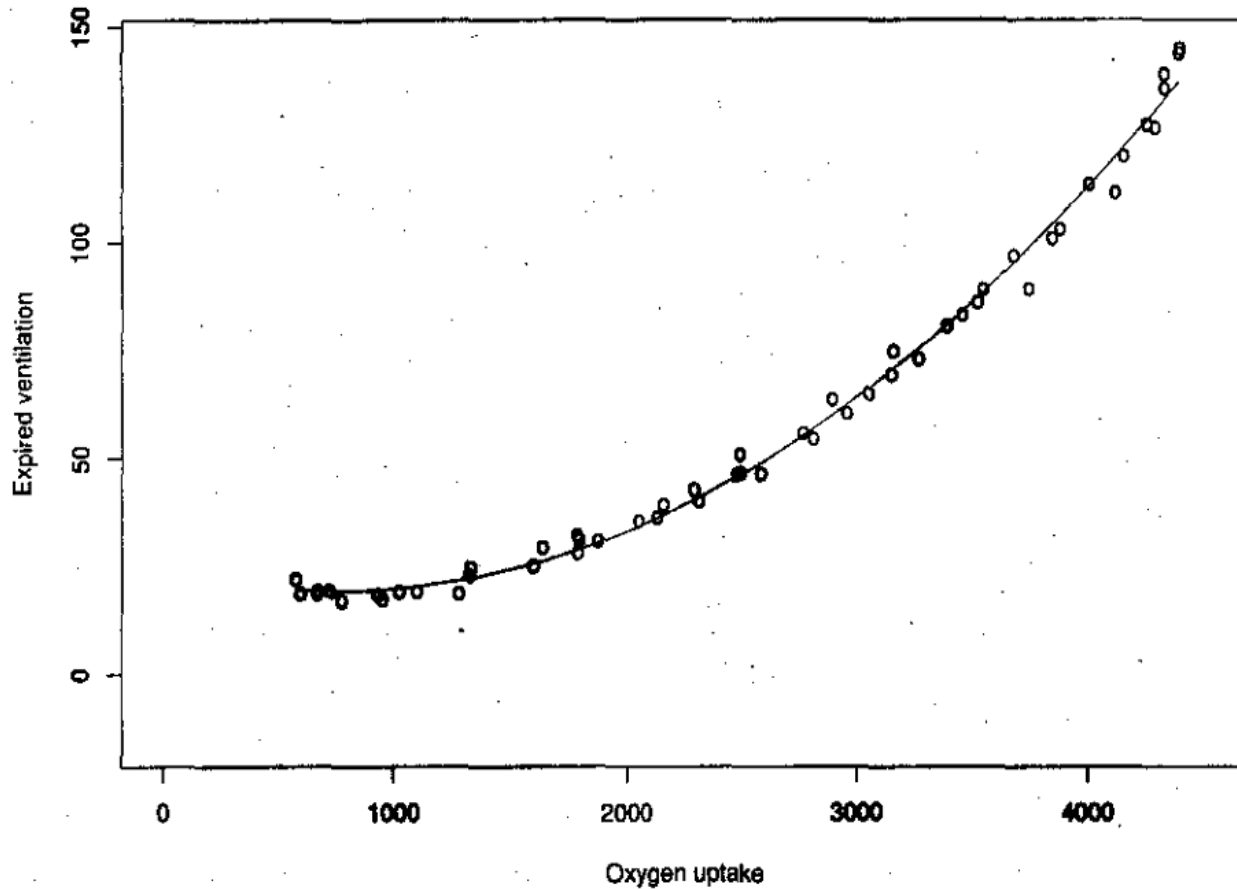


Figure 11.2, [2], pp. 373. The data from Figure 11.1 with a model that includes a term in x^2 : $y = \theta_0 + \theta_1x + \theta_2x^2$.

2.2. MULTIVARIATE LINEAR REGRESSION

- The house prices is affected by several variables/factors

Size (feet ²)	Number of rooms	Floors	Age	Price(\$1K)
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178
...	...	...	...	...

n: number of input attributes (e.g., n = 4)

$x^{(i)}$: input (features) of the i^{th} training sample

$x_j^{(i)}$: value of attribute j in the training sample i^{th}

$y^{(i)}$: i^{th} output in the training dataset

E.x.,

$$x^{(1)} = \begin{bmatrix} 2014 \\ 5 \\ 1 \\ 45 \end{bmatrix}; x_3^{(1)} = 1$$

2.2. MULTIVARIATE LINEAR REGRESSION

- Hypothesis:

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n$$

$$h_{\theta}(x) = 100 + 3x_1 + 2x_2 + 1.5x_3 - 2x_4$$

- Presenting in a matrix form ($x_0=1$)

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}; \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix}; \theta^T = [\theta_0, \theta_1, \theta_2, \dots, \theta_n]$$

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n = \theta^T x$$

2.2. MULTIVARIATE LINEAR REGRESSION

- Gradient descent:

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n = \theta^T x$$

- Coefficients $\theta(\theta_0, \dots, \theta_n)$: an $n+1$ vector

- Minimize: $J(\theta) = J(\theta_0, \dots, \theta_n)$

$$J(\theta) = \frac{1}{2N} \sum_{i=1}^N (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Repeat until convergence{

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta) \quad // \text{ simultaneously update for every } j=0, \dots, n$$

}

2.2. HỒI QUI TUYẾN TÍNH ĐA BIẾN

Repeat until convergence{

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta) \quad // \text{simultaneously update for every } j=0, \dots, n$$

}

$$\theta_0 = \theta_0 - \alpha \frac{1}{N} \sum_{i=1}^N (h_{\theta}(x^{(i)}) - y^{(i)}) x_0^{(i)} \quad \leftarrow x_0^{(i)} = 1$$

$$\theta_1 = \theta_1 - \alpha \frac{1}{N} \sum_{i=1}^N (h_{\theta}(x^{(i)}) - y^{(i)}) x_1^{(i)}$$

$$\theta_2 = \theta_2 - \alpha \frac{1}{N} \sum_{i=1}^N (h_{\theta}(x^{(i)}) - y^{(i)}) x_2^{(i)}$$

....

2.2. MULTIVARIATE LINEAR REGRESSION

- Feature scaling: to assure that all input features are in the same scale
 - E.g., $x_1 = \text{size (0 - 2000 feet}^2\text{)}$
 $x_2 = \text{number of rooms (1 - 5)}$

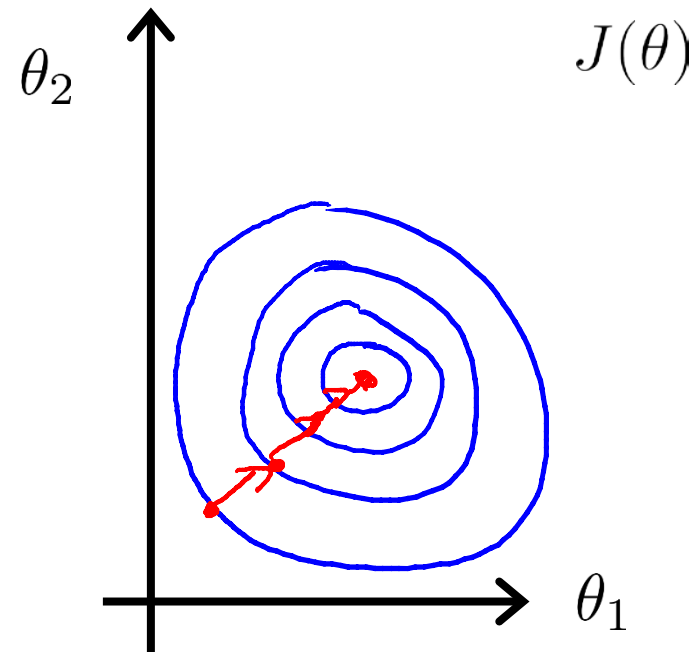
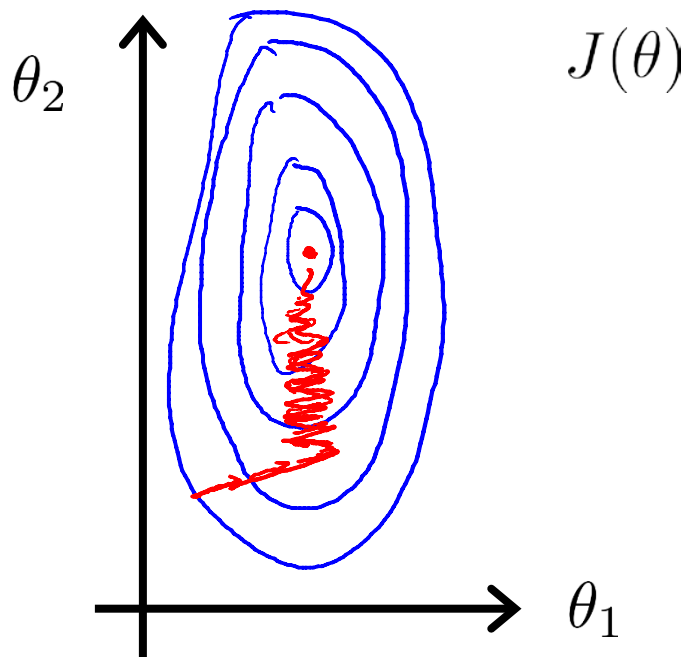
=> They are not in the same scale. The convergence speed is affected because of this imbalance scaling.

2.2. MULTIVARIATE LINEAR REGRESSION

- Assure that features are in the same scale

E.g. $x_1 = \text{size (0-2000 feet}^2\text{)}$
 $x_2 = \text{number of bedrooms (1-5)}$

Scaled: $x_1 = \text{size}/2000$
 $x_2 = \text{number of bedrooms} / 5$



2.2. MULTIVARIATE LINEAR REGRESSION

○ Feature scaling

- Normalize all feature to a range of $[-1, 1]$
- Ex., $x_0 = 1$; $0 \leq x_1 \leq 3$; $-2 \leq x_2 \leq 0.5$ $\Rightarrow$ **OK**
 $-100 \leq x_3 \leq 100$; $-0.0001 \leq x_4 \leq 0.0001 \Rightarrow$ **Normalize**

○ Apply normalization methods in chapter 2

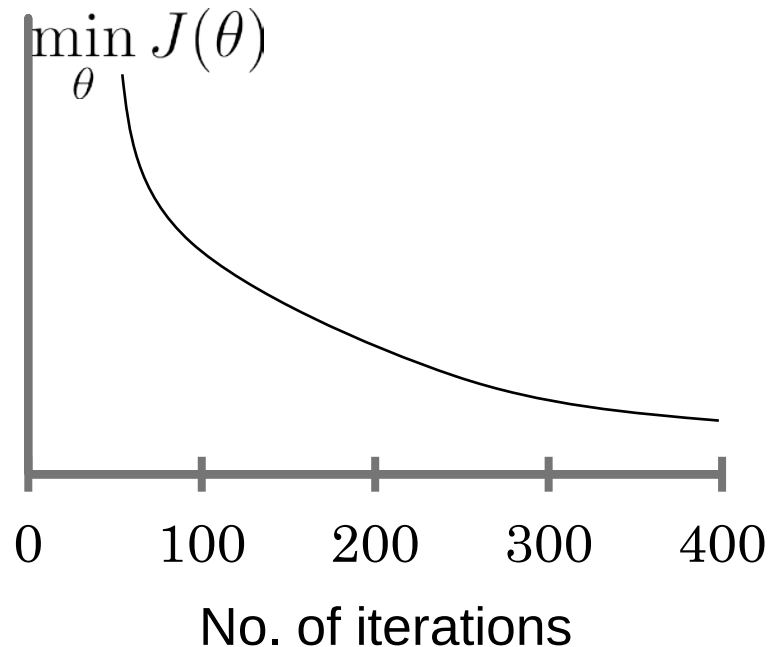
- Ex.

$$v' = \frac{v - \bar{v}}{\sigma v}$$

$$v' = \frac{v - \bar{v}}{V_{\max} - V_{\min}}$$

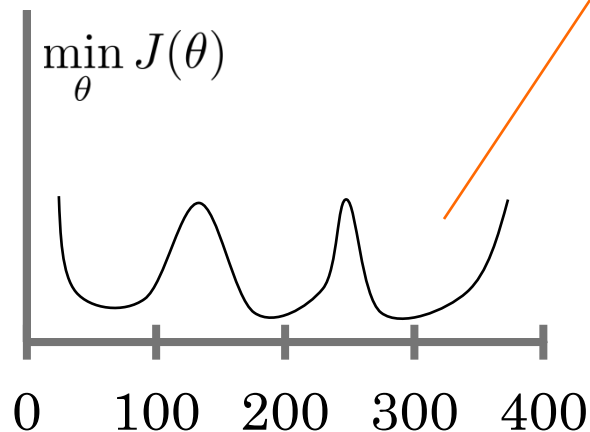
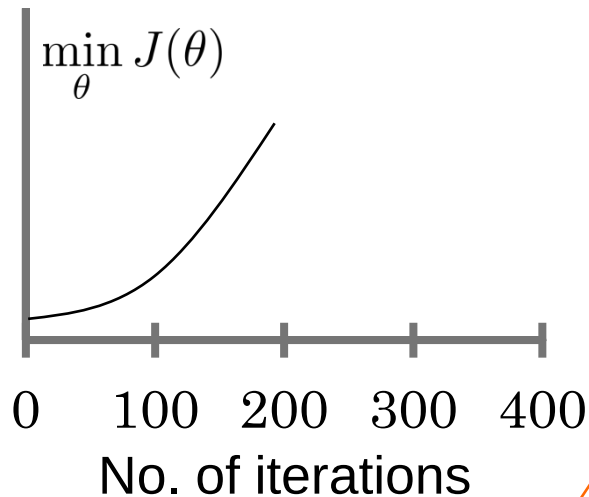
2.2. MULTIVARIATE LINEAR REGRESSION

- Validate the Gradient descent algorithm
 - $J(\theta)$ must decrease after each iteration
 - We can plot $J(\theta)$ by θ for intuitively check the convergence ability of the algorithm

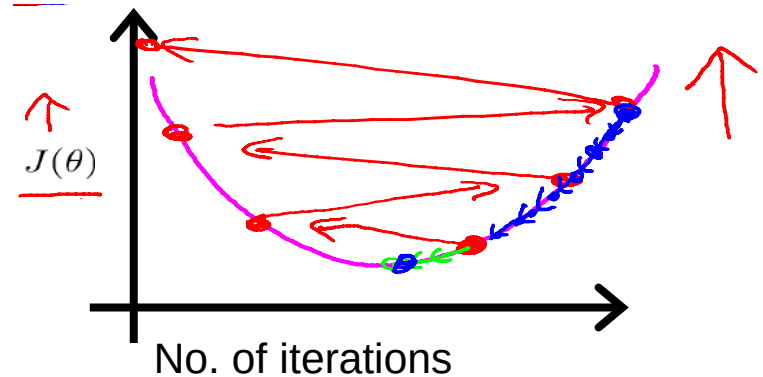


2.2. MULTIVARIATE LINEAR REGRESSION

Un-converged gradient descent



Should reduce α



- **α too small:** slow convergence
- **α too large:** $J(\theta)$ may not reduce at each iteration $\Rightarrow$ the algorithm may not converge
- **try α :** 0.001, 0.003, 0.01, 0.03, 0.1, 0.3, ...

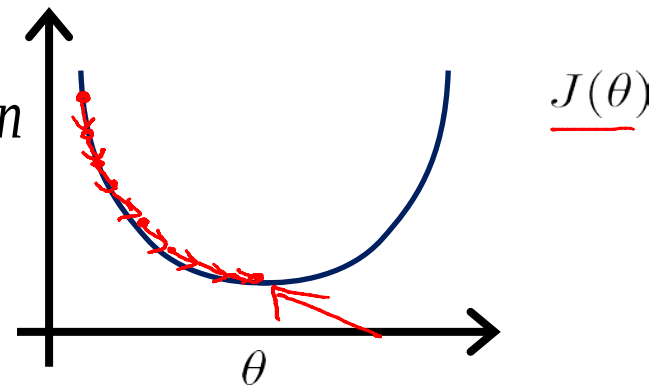
2.2. MULTIVARIATE LINEAR REGRESSION

- Use normal equation to identify θ

$$\frac{\partial}{\partial \theta_j} J(\theta) = \frac{1}{N} \sum_{i=1}^N (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)} = 0; \forall j = 1..n$$

$$\theta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

Gradient Descent



2.2. MULTIVARIATE LINEAR REGRESSION

Examples: $N = 4$, X : an $N \times (n+1)$ matrix; y : an $N \times 1$ matrix

x_0	size (feet ²) x_1	No. of rooms x_2	Floors x_3	Years x_4	Price (\$1K) y
1	2104	5	1	45	460
1	1416	3	2	40	232
1	1534	3	2	30	315
1	852	2	1	36	178

$$X = \begin{bmatrix} 1 & 2104 & 5 & 1 & 45 \\ 1 & 1416 & 3 & 2 & 40 \\ 1 & 1534 & 3 & 2 & 30 \\ 1 & 852 & 2 & 1 & 36 \end{bmatrix}$$

$$y = \begin{bmatrix} 460 \\ 232 \\ 315 \\ 178 \end{bmatrix}$$

$$\theta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

2.2. MULTIVARIATE LINEAR REGRESSION

- Given a training dataset: N examples, n features

Gradient descent

- must select α
- needs a large number of iterations
- workable event with a large n (e.g., $n = 10^6$)

Normal equation

- don't need to select α
- don't need iterations
- Must compute $(X^T X)^{-1}$
- May not work when n is large (when $n = 10^4$ then gradient descent should be used)

2.2. MULTIVARIATE LINEAR REGRESSION

- Note: the non-invertible issue in the normal equation method, i.e., $(X^T X)$ is not invertible
- Resolve:
 - Check the linear dependence of variables. Ex., the size in meter (x_1) and the size in feet (x_2) => remove dependent variables
 - Too much features ($n > N$). Ex., $n = 20$, $N = 10$ => reduce the number of features; find surrogate features; correct more data samples,...

2.2. MULTIVARIATE LINEAR REGRESSION

- Another example:

Quantity Sold	Price(\$)	Advertising (\$)
8500	2	2800
4700	5	200
5800	3	400
7400	2	500
6200	5	3200
7300	3	1800
5600	4	900

2.2. MULTIVARIATE LINEAR REGRESSION

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.980681
R Square	0.96174
Adjusted R Square	0.942604
Standard Error	310.5239
Observations	7

$$y = \text{Quantity Sold} = 8536.214 - 835.722 * \text{Price} + 0.592 * \text{Advertising}$$

ANOVA

	df	SS	MS	F	Significance F
Regression	2	9694300	4847150	50.26854	0.0014641
Residual	4	385700.4	96425.11		
Total	6	10080000			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	8536.21	386.9117	22.06243	2.5E-05	7461.974654	9610.453	7461.975	9610.453
Price(\$)	-835.72	99.65304	-8.38632	0.001106	-1112.40356	-559.041	-1112.4	-559.041
Advertising (\$)	0.59223	0.104347	5.675579	0.004755	0.302515325	0.881942	0.302515	0.881942

3. NON-LINEAR REGRESSION

○ $Y = f(X, \theta)$

- Y is a non-linear function in terms of relationship between parameters θ .
- Ex: Exponential, logarithmic function, Gauss, ...

$$f(x, \theta) = \frac{\theta_1 x}{\theta_2 + x}$$

○ Identify optimal θ : Optimization algorithms

- Local optimization
- Global optimization (using sum of squared residuals/errors)

4. APPLICATIONS

- Data mining
 - Data preprocessing: Smoothing, noise removal,...
 - Mining tasks: numerical-values prediction, descriptive analysis
- Apply in many domains: biology, agriculture, social issues, economy, business, finances, insurance, e-commerce, marketing, security, science, robotics, control systems, automation,...

5. ISSUES IN REGRESSION

- Assumptions
 - Data distribution: the relationship between predictors and dependent variables
 - Independence of predictors
 - Continuous values of variables (both predictor & responses)
 - Errors: How to identify them?
- The amount of data processed is not large
- How to identify the regression model
- Advanced techniques for regression:
 - Artificial Neural Network (ANN)
 - Support Vector Machine (SVM)

5. ISSUES IN REGRESSION

- Evaluation of a regression model:
 - Collect new data to evaluate the prediction results
 - Use the existing data (as testing dataset) for evaluation
 - Data splitting
 - ✓ Training data: To build the model
 - ✓ Testing data → *validate/evaluate the model*
 - K-fold cross-validation
 - ✓ Iterate k times:
 - ✓ Training data: (k-1) portions of data
 - ✓ Test data: the k^{th} portion of data → *accuracy*
 - ✓ *Average(accuracy) of k times*

5. ISSUES IN REGRESSION

○ Evaluation of the regression model:

- Accuracy
 - ✓ Sum of squared errors (SSE)
 - > Overall measure of errors: **smaller is better**

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- ✓ Mean squared error (MSE): measure of the variability in the response variable left unexplained by the regression: **smaller is better**

$$MSE = \frac{SSE}{n - m - 1}$$

(n: sample size, m: number of regression coefficients)

5. ISSUES IN REGRESSION

○ Accuracy (Con't)

- ✓ The standard error of the estimate (S)
- ✓ Đánh giá sai số thông thường trong quá trình dự đoán, sự sai lệch giữa giá trị dự đoán và giá trị thực của biến đáp ứng
- ✓ Measure the common error in the prediction process. It is the mean difference between the predicted and the actual values.
- ✓ Presents the precision of the prediction generated by the regression model

$$S = \sqrt{MSE} = \sqrt{\frac{SSE}{n - m - 1}}$$

5. ISSUES IN REGRESSION

- Factors affect the success of building regression models
 - ✓ Proper problem formulation
 - ✓ Selection of important variables and model form
 - ✓ Good dataset (both in volume and quality)
 - ✓ The use of good coefficient estimation procedures (e.g., gradient descent)
 - ✓ Model validation techniques

6. SUMMARY

- Regression
 - A statistical technique, applied to continuous attributes/features
 - Simple yet useful, applicable in various domains
 - One of example showing the contribution of statistics in data mining
- Types: Linear/non-linear, Univariate/Multivariate, Parametric/Non-parametric/Semi-parametric, Symmetric/Assymmetric

Q&A

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